DESIGN AND EVALUATION OF A GRAPH-BASED BOUNDED RISK PROPAGATION MODEL FOR COMPARATIVE PRIORITIZATION ACROSS DIRECTED-NETWORK DOMAINS

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CHAPTER I
INTRODUCTION

1.1 Background of the Study

Urban drainage networks are essential infrastructure for managing stormwater and reducing rainfall-induced flooding. Because drainage components are physically connected and flow is directional, the condition of one component can have consequences beyond the location where the disturbance is first observed. Studies on drainage malfunction, clogging, and blockage show that impaired components can reduce effective conveyance, alter flood behavior, and contribute to wider system consequences rather than remaining purely local events (Haghbin and Mahjouri, 2026), (Funke et al., 2025; Iqbal and Riaz, 2024; Wang et al., 2025). This makes drainage assessment a network problem rather than only a collection of isolated local inspections.

Previous drainage studies already demonstrate the practical value of prioritization. Complex-network and graph-based methods have been used to identify critical pipes, vulnerable components, weak links, and high-risk pathways by combining network structure with information such as hydraulic relevance, runoff, capacity, storage, discharge, failure combinations, or siltation condition (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). At the same time, computer-vision and monitoring approaches can identify local blockage, debris, or component condition at individual assets (Vandaele et al., 2024; Iqbal et al., 2021; Patil et al., 2023). These two research directions address different parts of the decision problem. Local monitoring can show how impaired a component is at the point of observation, while graph-based approaches can show where a component is structurally or functionally important. The literature therefore supports the distinction between local condition information and network-context prioritization rather than treating them as interchangeable outputs (Simone et al., 2022), (Dastgir et al., 2023), (Meijer et al., 2023; Zhang et al., 2025; Vandaele et al., 2024; Iqbal et al., 2021; Patil et al., 2023).

This distinction is important for inspection and maintenance decisions. In a network containing many connected components, decision-makers may need to determine which locations deserve earlier inspection, maintenance, or higher-fidelity analysis. Drainage studies on critical-pipe identification, weakest-link analysis, siltation chains, and multiple-pipe failure show that component importance can change when flow paths, capacity, network structure, or interacting failures are considered (Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). Consequently, two components with similar local disturbance may differ in network-level importance, while a structurally central component may not necessarily carry the highest current modeled risk under the present disturbance and operating conditions. The practical need addressed by this study is therefore not simply to detect blockage or rank structural importance, but to compare components according to a risk state that incorporates both local and propagated

network context.

The broader propagation literature shows that this modeling issue extends beyond drainage. Influence-diffusion models propagate activation, epidemic models propagate compartment or infection states, cascading-failure models propagate load or failure, attack-graph and cyber-risk approaches propagate reachable compromise or risk, and financial and infrastructure models propagate exposure, loss, stress, or related system states (Kempe et al., 2003; Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021; Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). These model families demonstrate that network propagation is well established, but they also show that the quantity being propagated differs substantially across applications. The update and aggregation rules used in each model are therefore tied to the semantics of that state rather than being automatically transferable to a bounded comparative risk-index task (Kempe et al., 2003), (Buldyrev et al., 2010), (Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018).

Across these model families, several recurring mechanisms are nevertheless evident. Topology establishes where interaction may occur; transmission or influence represents how strongly an effect transfers; susceptibility, capacity, vulnerability, or tolerance modifies how a relation or receiving context responds; current state influences later propagation; and incoming contributions require an aggregation rule consistent with the meaning of the propagated quantity (Kempe et al., 2003), (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). Drainage studies reinforce the same distinction by showing that structural measures are often augmented with hydraulic relevance, capacity, runoff, storage, discharge, or other operating information when functional importance is being assessed (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). These mechanisms provide useful building blocks, but the reviewed evidence does not clearly establish their particular integration into a lightweight directed-graph architecture whose primary output is a bounded continuous non-probabilistic comparative risk state.

The present study therefore proposes the **Graph-Based Bounded Risk Propagation Model (GBBRPM)**. The term bounded describes a core property of the model rather than a drainage-specific disturbance: the model represents node risk as a normalized state constrained to the interval [0,1]. The architecture is generic in formulation, while its parameter definitions remain domain-specific. In a drainage instantiation, for example, local disturbance may represent blockage severity and susceptibility may be derived from load relative to effective capacity. A different domain may use the same core only if it can defensibly define its own nodes, directed relations, disturbance, susceptibility, transmission, and comparative risk semantics. The study therefore claims generic formulation rather than universal empirical generalizability.

1.2 Statement of the Problem

The study addresses the problem of *comparative risk prioritization in interconnected directed networks*, initially motivated by urban drainage. Existing detection and monitoring approaches can identify the condition of an individual drainage component, including visual blockage or obstruction, but these outputs are principally local to the monitored asset (Vandaele et al., 2024; Iqbal et al., 2021; Patil et al., 2023). A local blockage score may therefore indicate that a component is impaired without directly showing how that condition should be prioritized once upstream influence, downstream connectivity, or relation-specific operating conditions are considered.

Graph-based structural approaches provide an important second perspective because they use network topology to identify critical or vulnerable components. However, drainage studies show that structural importance and functional or operating response are not equivalent. Simone et al. and Reyes-Silva et al. show that structural centrality may diverge from hydraulically meaningful importance, while later critical-pipe and weakest-link studies augment graph structure with runoff, capacity, storage, discharge, or other domain information (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023). Siltation-risk analysis similarly combines centrality, pipeline risk, edge vulnerability, and propagation pathways rather than relying on topology alone (Zhang et al., 2025). Centrality can therefore identify structurally important locations, but it does not by itself model how a current disturbance recursively changes as it moves through relation-specific operating conditions.

Existing propagation models address network influence from other directions, but their state semantics are commonly designed for different analytical questions. Influence-diffusion models use activation states (Kempe et al., 2003), epidemic models use infection or compartment states (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021), cascading-failure models use load, capacity, or failure states (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017), attack-graph approaches use reachable compromise or probability (Sheyner et al., 2002), (Muñoz-González et al., 2017), and cyber, finance, transportation, and infrastructure models use other forms of risk, exposure, loss, or dynamic system state (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). These formulations are appropriate for their intended purposes, but they do not directly provide the bounded continuous non-probabilistic comparative risk state targeted in the present study.

Taken together, the literature reveals an architectural and integrative research problem. The individual mechanisms relevant to the target task already exist: directed reachability is established in graph models; recursive current-state updating appears across contagion and cascading processes; capacity, vulnerability, susceptibility, or related response factors appear in infrastructure and risk models; and multiple forms of aggregation are used according to state semantics (Kempe et al., 2003; Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021; Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et

al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). Drainage research additionally demonstrates the practical importance of combining structural and operating information for prioritization (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). Within the reviewed evidence, however, these mechanisms are not clearly consolidated into one lightweight and interpretable architecture specifically intended to produce a bounded comparative risk state from local disturbance and propagated network context.

The central problem of the study is therefore how to formulate a directed-graph model that combines local disturbance with recursively propagated network context so that nodes can be compared according to their current modeled risk state, while keeping topology, susceptibility, optional transmission, aggregation, and dependence conceptually distinct. Consistent with this central problem, the study addresses the following research questions:

1. How can local disturbance, current upstream risk, relation-specific susceptibility, optional transmission, directed reachability, and bounded multi-source aggregation be integrated into an interpretable directed-graph architecture?
2. Does the resulting formulation preserve boundedness, monotonicity, and expected comparative-ranking behavior under controlled network conditions?
3. How sensitive, robust, and computationally scalable is the model under changes in disturbance severity, susceptibility, topology, disturbance location, and multi-source configuration?
4. Can the same bounded propagation architecture be instantiated coherently in drainage, software-dependency, and electrical-network cases without redefining its core update rule?
5. What evidence does each domain provide for model behavior, structural applicability, preprocessing feasibility, and the boundary between comparative graph risk and domain-specific outcome prediction?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of the study is to design and computationally evaluate a **Graph-Based Bounded Risk Propagation Model (GBBRPM)** for interpretable comparative node-risk prioritization in directed networks and to examine its cross-domain applicability through drainage, software-dependency, and electrical-network cases.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. synthesize the limitations and recurring mechanisms of local condition assessment, graph-based prioritization, and existing propagation models that are relevant to comparative network-risk prioritization;

2. formulate a bounded continuous non-probabilistic recursive architecture that explicitly distinguishes local disturbance, relation-specific susceptibility, optional transmission, directed reachability, current-state updating, bounded multi-source aggregation, and dependence;
3. computationally validate the generic GBBRPM architecture using controlled and reconstructed networks in terms of boundedness, monotonicity, ranking behavior, sensitivity, robustness, reconvergence effects, and computational scalability;
4. instantiate the frozen architecture in drainage, software-dependency, and electrical-network cases using traceable topology, documented domain-specific parameter mappings, and explicitly labeled controlled sensitivity settings, including measured-loading electrical experiments; and
5. compare the resulting evidence across domains while distinguishing behavioral validation, structural applicability, baseline-data verification, and independent outcome validation, and identify the model's scope boundaries and requirements for further refinement.

1.4 Significance of the Study

The study is significant because it addresses a decision-support space between purely local or structural prioritization and high-fidelity domain simulation. Drainage literature demonstrates both the usefulness of lightweight graph-based screening and the added physical detail provided by hydrodynamic and learned hydraulic models (Simone et al., 2022), (Dastgir et al., 2023), (Meijer et al., 2023), (Zhang et al., 2024). GBBRPM does not attempt to replace those higher-fidelity approaches. Instead, it is intended to produce a bounded comparative risk state that allows components to be ranked relative to one another under a common network and parameterization.

For urban drainage research, the study provides a way to connect local disturbance information with network context for preliminary prioritization of locations that may merit inspection, maintenance, or higher-fidelity hydraulic analysis. This positioning is consistent with drainage studies that use graph methods for critical-pipe or weakest-link screening while retaining hydrodynamic models for richer physical assessment (Dastgir et al., 2023), (Meijer et al., 2023). The model is therefore intended as a complementary prioritization layer rather than a hydraulic prediction model.

For graph-based risk and propagation research, the study provides an explicit architecture in which topology, susceptibility, transmission, local disturbance, aggregation, and dependence are treated as conceptually distinct mechanisms rather than being collapsed into one undifferentiated coefficient. This separation follows recurring distinctions observed across influence, cascade, cyber-risk, financial, and infrastructure propagation models (Kempe et al., 2003), (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017; Sheyner et al., 2002; Muñoz-González et al., 2017; Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018). For the drainage application and any future domain-specific application, the model provides a reusable

core propagation structure while requiring each instantiation to justify its own parameter definitions and any additional refinement layer.

The study is also significant methodologically because interpretability is retained throughout the propagation process. A high modeled risk state at a node can be traced to the current upstream state, relation-specific susceptibility, optional transmission, local disturbance, and the bounded aggregation mechanism. This traceability is especially relevant to comparative prioritization because the purpose is not only to produce an ordering, but also to explain why one component receives a higher modeled risk state than another.

1.5 Scope and Limitations

The study focuses on the design, computational validation, and cross-domain evaluation of GBBRPM as a lightweight directed-graph architecture for comparative bounded-risk propagation. The generic core represents a node risk state, a local disturbance, relation-specific susceptibility, optional transmission or influence, and an edge contribution. The risk state and the model's bounded parameters are represented on normalized [0,1] scales.

The evaluation has two complementary layers. The first evaluates the generic formulation on directed acyclic graphs using one-pass topological processing, controlled synthetic networks, diagnostic baselines, perturbation trials, mathematical property tests, and scalability tests. The second examines cross-domain instantiation through drainage, software-dependency, and electrical-network cases. Drainage supplies behavioral tests and a SWMM external-reference comparison; software supplies real-topology structural applicability and transmission sensitivity evidence; and electrical data supply real-topology and baseline-preprocessing verification together with controlled mechanism and sensitivity tests on an observable Component 2 model parameterized from measured loading. Event-centered electrical outcome validation remains pending because the available measurements represent normal operation and are not aligned with independent outcomes for the documented switching events.

The study does not interpret the bounded risk state as a calibrated probability of failure, blockage, or any other event. The base model also does not represent cycles, bidirectional feedback, explicit temporal dynamics, hydraulic backwater, surcharge, ponding, flow redistribution, or common-cause dependence correction. These exclusions are important because drainage and broader propagation literature show that feedback, shared dependencies, and richer physical state interactions may require iterative, probabilistic, or high-fidelity formulations beyond a one-pass lightweight graph model (Buldyrev et al., 2010), (Shuang et al., 2017), (Wu et al., 2026), (Petrone and Latora, 2018), (Dastgir et al., 2024), (Zhang et al., 2024). Where heterogeneous transmission cannot yet be empirically justified, the baseline transmission term is kept neutral. Likewise, drainage-specific susceptibility mappings such as load relative to effective capacity are treated as instantiation choices rather than universal components of the generic model.

The study claims generic formulation and demonstrated cross-domain instantiability within the three evaluated cases, not universal empirical gen-

eralizability. The cases test different claims and do not provide equivalent outcome-validation strength. A domain may instantiate the core only if it can provide defensible definitions for nodes, directed relations, local disturbance, susceptibility, optional transmission, and comparative risk. Domain-specific prediction, calibration, and operational usefulness must be justified and validated separately from the generic core.

1.6 Conceptual Framework

The conceptual framework of the study begins with local disturbance and directed network context. The current risk state at an upstream node is transformed into a relation-specific contribution according to susceptibility and, where justified, optional transmission. Multiple incoming contributions are then combined with any local disturbance at the receiving node using a bounded aggregation rule, producing an updated comparative risk state. That updated state can in turn influence later downstream relations, allowing the modeled risk state to evolve recursively through the network. The resulting node states provide the basis for comparative prioritization.

Figure 1.1 distinguishes the domain-supplied inputs, the generic bounded propagation core, and the comparative decision-support output.

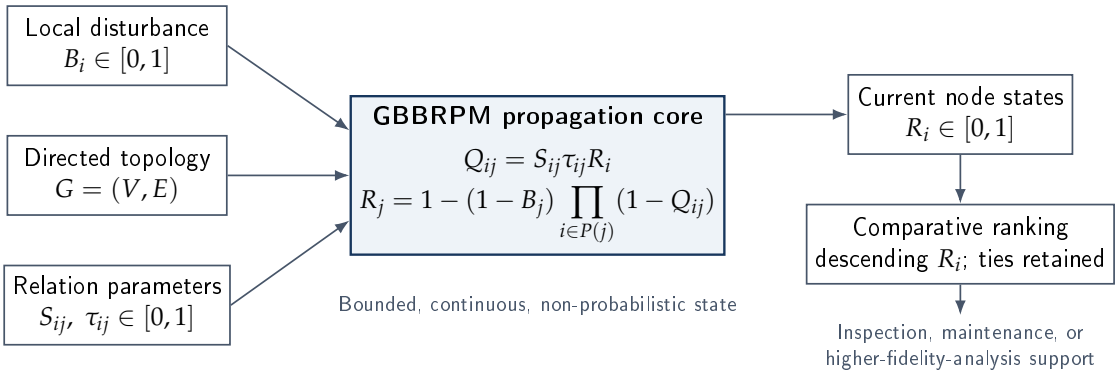


Figure 1.1. Conceptual framework of GBBRPM, distinguishing supplied domain inputs, the generic propagation core, and the comparative decision-support output.

For a directed relation $i \to j$, the generic edge contribution is:

$$Q_{ij} = S_{ij}\tau_{ij}R_i \qquad \textit{(Equation 1-1)}$$

where R_i is the current upstream normalized risk state, S_{ij} is relation-specific susceptibility, and τ_{ij} is optional transmission or influence.

The receiving-node state is:

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}) \qquad \textit{(Equation 1-2)}$$

where B_j is the local disturbance and $P(j)$ is the set of predecessors of node j. The multiplicative-complement form is used as a bounded saturating aggregation mechanism. Although it is algebraically similar to noisy-OR, it is not interpreted as a probabilistic union because the inputs are bounded risk contributions rather than calibrated probabilities.

The resulting R_j is the node's current modeled risk state. It may then serve as the upstream state for later relations, allowing risk to be recursively updated through the network. Node ranking is therefore a decision-support output derived from the resulting risk states rather than the propagation mechanism itself.

1.7 Definition of Terms

Term	Definition
Comparative risk prioritization.	Ranking components according to their modeled risk states relative to one another under the same network and parameterization.
GBBRPM.	Graph-Based Bounded Risk Propagation Model, the generically formulated directed-graph architecture proposed in this study.
Normalized risk state.	A dimensionless continuous value in $[0, 1]$ representing the current modeled level of network-context risk at a node. It is not a calibrated probability of an event.
Local disturbance.	A bounded input representing risk or disturbance introduced directly at a node according to the selected domain.
Susceptibility.	A bounded relation-specific factor representing how strongly a relation or receiving context responds under the evaluated condition.
Transmission or influence.	An optional bounded factor describing how strongly influence transfers along an existing directed relation.
Edge contribution.	The relation-specific contribution generated from current upstream risk, susceptibility, and optional transmission.
Topology or reachability.	The directed network structure that determines which nodes and relations can participate in propagation.
Bounded aggregation.	The multiplicative-complement mechanism used to combine local disturbance and multiple incoming contributions while preserving the $[0, 1]$ state range.

Term	Definition
Domain-specific instantiation.	The process of defining the generic model's nodes, edges, disturbance, susceptibility, transmission, and any optional refinement parameters according to a particular application domain.
Generic formulation.	A model architecture whose core propagation logic is not hard-coded to one application domain. Generic formulation does not imply that the model has been empirically evaluated across all domains.

CHAPTER II
REVIEW OF RELATED LITERATURE

2.1 Review Basis and Organization

This chapter synthesizes two complementary evidence streams. The first concerns urban drainage monitoring, blockage, graph-based vulnerability assessment, critical-component analysis, and hydraulic or network-level prioritization. This literature establishes the practical origin of the research problem. The second concerns foundational and modern propagation models across influence diffusion, epidemics, cascading failure, cybersecurity, finance, transportation, and infrastructure. This literature establishes the mechanisms and state semantics that can inform a general graph-risk architecture.

The review follows a two-stage synthesis. It first establishes the domain-specific problem through drainage monitoring and graph-based prioritization. It then situates that problem within the broader literature on propagation-state semantics, topology, transmission, susceptibility, recursion, aggregation, dependence, model complexity, and ranking. The final sections compare reusable architectures and domain-specific parameterization before synthesizing the research problems and the architectural gap addressed by the study.

2.2 Urban Drainage Networks as Interconnected Systems

Urban drainage networks consist of connected conduits, junctions, inlets, screens, and other components whose behavior is influenced by flow direction, network structure, hydraulic condition, and local impairment. Blockage, clogging, siltation, structural defects, or reduced effective capacity can influence system behavior beyond the initially affected location. Numerical and network-based drainage studies show that the location and severity of impairment can alter hydraulic response, vulnerability, and the importance of connected components (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

This interconnected structure creates a practical prioritization problem. Inspection and maintenance resources are finite, so not every component can receive equal attention at the same time. Existing drainage studies explicitly frame critical-component identification as useful for targeted management, maintenance, monitoring, resilience, or dredging decisions (Simone et al., 2022), (Simone, 2023), (Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025). The practical decision problem is therefore not only to identify which components are locally impaired but also to determine which components deserve greater attention when the surrounding network is considered.

This drainage-side evidence exposes a distinction between observation and prioritization. A local blockage or defect can be observed directly, but a maintenance decision often requires a broader question: whether that component

remains important after the connected network, downstream pathways, relation conditions, and converging influences are considered. This distinction between local condition and network-context priority recurs in studies that combine inspection, topology, capacity, and hydraulic relevance (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.3 Local Monitoring and Component-Level Condition Assessment

Computer-vision, sensor, and inspection studies have substantially improved the ability to identify local drainage condition. Iqbal et al. (2021), for example, formulate culvert blockage as an image-classification problem and evaluate several convolutional neural networks for classifying blockage from imagery. Vandaele et al. (2024) similarly develop camera-based methods for automatically detecting blockage at trash screens for maintenance and flood-management use. These approaches demonstrate that local blockage information can be generated automatically and repeatedly at monitored assets.

The outputs of these methods are valuable because they provide repeatable local condition or disturbance signals. However, a local score by itself does not determine network-context priority: it describes the monitored asset rather than the recursively accumulated state of the wider connected system (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022).

This distinction matters when two components have similar observed local conditions but occupy different positions in the network. One may lie on a relation with high operating susceptibility or feed several important downstream paths, while another may have little propagated consequence under the same network configuration. Local assessment is therefore necessary for condition awareness, but it does not by itself answer which nodes warrant greater priority after network position and operating conditions are considered (Simone et al., 2022; Dastgir et al., 2023; Zhang et al., 2025).

The limitation should not be interpreted as a weakness of computer vision or sensing. Those methods solve the detection problem they were designed to solve. The unresolved issue is the interface between local condition information and network-level prioritization. This motivates the need for a propagation layer that can accept a local disturbance as input while separately representing network structure and relation-specific response (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023).

2.4 Graph-Based Drainage Analysis and Structural Prioritization

Drainage research has increasingly used graph theory and complex-network analysis to identify critical or vulnerable components. Simone et al. (2022) examine standard and tailored centrality metrics for vulnerability assessment, monitoring design, and contaminant or pathogenic spread. Their analysis shows that standard degree, harmonic, betweenness, and edge-betweenness measures do not always reproduce hydraulically meaningful importance, which motivates tailored metrics that incorporate intrinsic relevance and flow direction.

Simone (2023) extends this idea using relevance-based centrality for critical-pipe identification. The method ranks nodes and pipes by combining network connectivity with intrinsic relevance rather than using plain topological centrality alone. This is significant for the present study because it demonstrates that even within a graph-only framework, useful prioritization can require information beyond raw connectivity.

Dastgir et al. (2023) propose a graph method for critical-pipe analysis based on runoff edge betweenness and capacity-related information. Their method is compared with hydrodynamic simulation and is intended to retain useful criticality information with lower computational and data requirements. The study also reports that graph-method accuracy decreases as network complexity, loop degree, and hydraulic stress increase, illustrating the tradeoff between lightweight structural approximation and richer physical behavior.

Dastgir et al. (2024) subsequently examine multiple-pipe failures in stormwater networks using graph-theoretic representations. Their work is particularly relevant to the present study because it shows that the consequences and prioritization of interacting failures cannot always be inferred from isolated single-component analysis. It therefore supports the explicit evaluation of multi-source configurations and converging effects while also reinforcing the need to distinguish graph-based screening from full hydraulic consequence prediction.

Meijer et al. (2023) develop a graph-based weakest-link method for urban drainage systems using flow-path analysis instead of repeated hydrodynamic simulation. Their work focuses on storage and discharge-capacity relationships and demonstrates that graph-based screening can make long rainfall-series analysis practically feasible. This provides further evidence that drainage research already seeks computationally efficient prioritization and vulnerability tools rather than relying exclusively on full hydraulic simulation.

Zhang et al. (2025) analyze siltation risk using complex-network concepts, centrality, pipeline risk factors, and edge vulnerability. Their study identifies critical nodes and high-risk propagation pathways, explicitly linking siltation-chain structure with maintenance and dredging prioritization. This is especially relevant because it shows that drainage risk prioritization may involve both structural importance and domain-specific condition information rather than a single metric.

Taken together, these studies establish two important points. First, prioritization is already a legitimate drainage-network task; the novelty of the present study is therefore not the act of ranking nodes or pipes. Second, the literature repeatedly augments topology with runoff, hydraulic relevance, storage/discharge capacity, edge vulnerability, siltation condition, or other operational information. Structural position and operating response are therefore related but not equivalent (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.5 Propagation Models and State Semantics

Broader network science provides several foundational propagation paradigms. Independent Cascade and Linear Threshold models represent influence through activation semantics (Kempe et al., 2003). Epidemic models such as SIR and SIS propagate compartment or infection states (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021). Cascading-failure models propagate physical load, capacity exceedance, or component failure (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017). Attack graphs represent feasible attack paths and reachable compromise states, while Bayesian attack-graph approaches introduce probabilistic compromise semantics (Sheyner et al., 2002), (Muñoz-González et al., 2017). Cyber-risk, finance, transportation, and infrastructure models introduce additional states such as accumulated risk, exposure, payment loss, systemic stress, or dynamic load (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026; Petrone and Latora, 2018).

The main lesson is that the meaning of the propagated state determines the valid update and aggregation rules. Binary activation answers whether a node changes state. Probability estimates event likelihood under a probabilistic model. Physical-load models track conserved or redistributed quantities. These states are appropriate for their intended questions, but they do not automatically answer the comparative question of which node carries greater comparative network-context risk under a common parameterization (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Kavallieratos et al., 2021).

The comparison indicates that any proposed propagation model must declare whether its state represents activation, probability, physical quantity, or a non-probabilistic index. Without that declaration, similarly bounded values may be interpreted incorrectly and rankings from different model families may appear comparable when they are not (Kempe et al., 2003; Kermack and McKendrick, 1927; Muñoz-González et al., 2017; Petrone and Latora, 2018).

Table 2.1 compares the model families at the level emphasized during consultation: their structure, propagated state, update basis, analytical purpose, and limitation for a lightweight comparative-risk task. The comparison shows why no single family can be adopted merely because it uses a graph; the semantics of its state and the assumptions of its update rule must match the research question.

Table 2.1. Comparison of propagation-model families relevant to comparative network-risk prioritization.

Model family	Typical state	Propagation or update basis	Principal analytical use	Limitation for the present problem
Influence diffusion (Kempe et al., 2003)	Binary activation or threshold state	Edge influence probabilities or accumulated neighbor influence	Spread maximization and activation reach	Activation semantics do not directly represent a continuous comparative risk index.
Epidemic propagation (Kermack and McKendrick, 1927; Pastor-Satorras and Vespignani, 2001; Humphries et al., 2021)	Susceptible, infected, and recovered compartments or infection probabilities	Contact structure, infection rate, and recovery rate	Disease spread and intervention analysis	Epidemiological compartments and rates require domain-specific assumptions not implied by network priority.
Cascading failure (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017)	Load, capacity, operational state, or failure state	Load redistribution, tolerance, and interdependence	Robustness and cascade consequence	Physical load conservation or failure thresholds are not generic comparative-risk semantics.
Attack graph and Bayesian attack graph (Sheyner et al., 2002; Muñoz-González et al., 2017)	Reachable compromise state or compromise probability	Feasible attack steps, conditional probabilities, or logical relations	Security-path and compromise analysis	Probabilistic outputs require calibrated conditional assumptions and dependence treatment.
Financial and infrastructure contagion (Faraji et al., 2026; Zhou et al., 2023; Wu et al., 2026)	Exposure, loss, stress, or service degradation	Dependency, exposure, vulnerability, and feedback	Systemic-risk and resilience assessment	Feedback and common exposure can require cyclic or dependence-aware dynamics.

Table 2.1. Comparison of propagation-model families relevant to comparative network-risk prioritization (continued).

Model family	Typical state	Propagation or update basis	Principal analytical use	Limitation for the present problem
Graph-based drainage screening (Simone et al., 2022; Simone, 2023; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025)	Centrality, criticality, capacity, or vulnerability score	Topology combined with relevance, runoff, capacity, or siltation information	Monitoring, maintenance, and critical-component screening	Scores differ in meaning and may omit recursive current-state propagation.
Hydraulic simulation and learned surrogate (Reyes-Silva et al., 2020; Dastgir et al., 2024; Zhang et al., 2024)	Flow, depth, surcharge, flood volume, or learned hydraulic state	Governing physics or simulator-trained approximation	Physical consequence analysis and prediction	Greater data, calibration, or training requirements; outputs answer a different question from lightweight comparative risk.

2.6 Topology, Reachability, and Transmission

Across propagation models, topology primarily determines which interactions are possible. An edge $i \rightarrow j$ indicates that influence may move from node i to node j, but the existence of the edge does not by itself define the strength of influence. Influence-diffusion models, attack graphs, cyber-risk models, and weighted infrastructure models commonly distinguish the existence of a pathway from the weight, probability, exposure, or transfer mechanism associated with that pathway (Kempe et al., 2003; Sheyner et al., 2002; Kavallieratos et al., 2021; Faraji et al., 2026; Petrone and Latora, 2018).

This distinction is particularly important for interpreting centrality. Centrality uses global graph structure and can identify structurally important nodes or edges, but it does not directly simulate the propagation of a current disturbance under current relation-specific conditions. Simone et al. (2022) provide a drainage-specific example: standard centrality metrics can rank structurally central elements differently from hydraulically relevant elements, which motivated the incorporation of relevance and direction.

Accordingly, the literature supports treating topology as reachability

while retaining transmission or influence as a conceptually separate relation-level mechanism. Where heterogeneous influence cannot be empirically justified, a neutral coefficient is more defensible than an unsupported set of edge weights (Kempe et al., 2003; Simone et al., 2022; Dastgir et al., 2023).

2.7 Susceptibility, Capacity, Vulnerability, and Receiver Response

A second distinction concerns transmission versus susceptibility. Transmission describes how strongly an influence transfers along a relation, whereas susceptibility describes how strongly a relation or receiving context responds under the current condition. Similar distinctions appear in models using capacity, vulnerability, tolerance, resilience, exposure, or fragility (Buldyrev et al., 2010; He et al., 2019; Kavallieratos et al., 2021; Faraji et al., 2026).

Cascading-failure models demonstrate that the same incoming load can produce different outcomes depending on component capacity or tolerance (Buldyrev et al., 2010; He et al., 2019; Shuang et al., 2017). Cyber and infrastructure models likewise separate dependency or exposure strength from vulnerability or resilience (Kavallieratos et al., 2021; Faraji et al., 2026; Zhou et al., 2023). Drainage studies that incorporate hydraulic relevance, runoff, capacity, storage/discharge relationships, siltation condition, or edge vulnerability similarly show that structural connectivity alone is not sufficient to characterize operating response (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

These patterns support keeping relation-specific susceptibility distinct from transmission rather than collapsing both concepts into one unexplained coefficient. In a drainage application, susceptibility may be linked to load relative to effective capacity, whereas heterogeneous transmission should remain neutral unless empirical evidence justifies its calibration (Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.8 Recursive Propagation and Current-State Updating

Many propagation systems are recursive: later effects depend on the state produced by earlier propagation rather than repeatedly referencing only the original source. Epidemic models update infection state over successive interactions, cascading-failure models redistribute load after each failure, financial-contagion models update exposure or loss as conditions change, and hydraulic models update system state in response to preceding conditions (Kermack and McKendrick, 1927; Buldyrev et al., 2010; He et al., 2019; Faraji et al., 2026; Petrone and Latora, 2018).

This pattern supports current-state recursion: for a chain $A \rightarrow B \rightarrow C$, the contribution from B to C should be based on the current modeled state at B when the intended semantics permit inherited and local influence to accumulate. Conceptually, the node propagates what it currently carries rather than only the disturbance that originally entered the network (Kermack and McKendrick, 1927; Buldyrev et al., 2010; Faraji et al., 2026).

Current-state recursion is central to comparative network context. If the model repeatedly used only the original local disturbance, intermediate accumulation would be lost and downstream ranking would not reflect the evolving state of the network (Kempe et al., 2003; Buldyrev et al., 2010; Petrone and Latora, 2018).

2.9 Multi-Source Aggregation and Bounded State

When several upstream relations converge on the same node, their contributions require aggregation. Existing model families use different operators according to state semantics: physical loads may be summed or redistributed, threshold models accumulate influence until a threshold is crossed, and probabilities may be combined through Bayesian or noisy-OR structures when the required assumptions are satisfied (Kempe et al., 2003; Muñoz-González et al., 2017; Buldyrev et al., 2010).

For a bounded non-probabilistic comparative risk state, simple addition creates a practical difficulty because multiple contributions can exceed the declared [0,1] range. Clipping can restore the range but introduces an abrupt ceiling that loses information about how additional contributions accumulate. Multiplicative-complement aggregation is a candidate bounded, saturating operator, although its resemblance to noisy-OR algebra requires an explicit warning: it has a probabilistic interpretation only when the inputs and dependence assumptions justify one (Muñoz-González et al., 2017). The selected operator must therefore follow the declared state semantics rather than visual similarity to an established probability formula.

2.10 Dependence, Shared Paths, and Reconvergence

Bounded aggregation does not solve every multi-source problem. When contributions share a common origin and travel through multiple paths that later reconverge, incoming contributions may not be independent descriptions of distinct risk sources. Similar dependence problems appear in financial networks through common exposures, in infrastructure networks through shared dependencies, and in graph systems with overlapping paths (Buldyrev et al., 2010; Faraji et al., 2026; Zhou et al., 2023).

The literature therefore supports treating local aggregation and dependence as separate mechanisms. Bounding incoming contributions does not by itself correct shared-origin or common-cause dependence; reconvergence should be evaluated as an explicit limitation rather than hidden inside the aggregation rule (Buldyrev et al., 2010; Faraji et al., 2026).

This distinction is important for interpretability. A bounded output does not automatically mean that shared-origin information has been counted correctly. Dependence-aware correction remains a separate extension problem (Muñoz-González et al., 2017; Faraji et al., 2026).

2.11 Lightweight Graph Models, Learned Surrogates, and High-Fidelity Simulation

The drainage literature contains a spectrum of model complexity. Local scores and graph indicators provide relatively inexpensive screening; graph-hydrodynamic hybrids combine structural reasoning with physical analysis; hydraulic simulators explicitly represent domain dynamics; and graph-neural-network surrogates learn simulator-derived hydraulic behavior for faster inference (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Dastgir et al. (2023) explicitly compare a graph-based critical-pipe method with hydrodynamic modeling and report substantially lower data and computational requirements for the graph approach, while also documenting accuracy loss as complexity and stress increase. Simone et al. (Simone et al., 2022) likewise position tailored complex-network metrics as complementary preliminary tools when full hydraulic simulation is more complex, data demanding, or computationally expensive.

At the learned-surrogate end of the spectrum, Zhang et al. (2024) develop a graph-neural-network surrogate for urban-drainage hydraulic prediction. The model is trained against physics-based flow-routing outputs and predicts hydraulic variables for real-time use. Physics-guided mechanisms are included to improve prediction consistency. Such a surrogate seeks to emulate hydraulic state transitions, which is a different objective from transparent graph screening or comparative index construction (Zhang et al., 2024).

The literature therefore does not imply that lightweight models are universally superior. High-fidelity models provide richer physical detail, while learned surrogates can approximate that detail after training. Lightweight graph models instead trade physical completeness for lower data requirements, simpler computation, and direct interpretability. The appropriate model depends on the decision question (Dastgir et al., 2023; Simone et al., 2022; Zhang et al., 2024).

2.12 Existing Ranking and Prioritization Approaches

Existing drainage and network studies rank components using different criteria. Local monitoring may rank according to observed blockage or defect severity. Centrality approaches rank according to structural importance. Weakest-link and vulnerability methods incorporate combinations of topology and domain-specific information. Failure studies rank components or combinations according to modeled consequence. Siltation studies identify high-risk nodes and propagation pathways. Hydraulic approaches may prioritize locations according to depth, surcharge, flood volume, resilience, or performance loss (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022; Dastgir et al., 2023, 2024; Meijer et al., 2023; Zhang et al., 2025).

The evidence shows that ranking is not a single modeling operation with a universal meaning. A centrality rank, blockage-severity rank, weakest-link rank, hydraulic-impact rank, and propagated-risk rank answer different questions. The semantic basis of the score must therefore be explicit before rankings can be

compared (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

Table 2.2. Comparison of ranking and prioritization approaches

Approach	Primary ranking basis	What the ranking answers	Limitation relative to the present target
Local condition / detection (Iqbal et al., 2021; Vandaele et al., 2024; Patil et al., 2023)	Observed blockage, defect, debris, or local score	Which asset appears more locally impaired?	Does not inherently incorporate recursively propagated network context
Centrality / structural criticality (Simone et al., 2022; Simone, 2023; Reyes-Silva et al., 2020)	Graph position, paths, relevance, or structural interaction	Which node or edge is more structurally important?	Does not by itself represent propagation of a current disturbance under current relation conditions
Weakest-link / vulnerability methods (Dastgir et al., 2023; Meijer et al., 2023)	Structural information combined with capacity, storage, discharge, redundancy, or other domain variables	Which component most constrains or weakens system performance?	Produces specialized vulnerability or criticality semantics rather than a generic bounded recursive risk state
Siltation-chain risk (Zhang et al., 2025)	Pipeline risk, node importance, edge vulnerability, and propagation path	Which nodes and pathways are high priority for siltation management?	Domain-specific siltation semantics and parameterization
Hydraulic consequence analysis (Dastgir et al., 2024; Zhang et al., 2024)	Simulated depth, surcharge, flood volume, performance loss, or resilience	Which failures or locations cause greater physical consequence?	High-fidelity but strongly domain-specific and more data/computation intensive

The literature comparison also separates score generation from the decision-support use of a score. A method may first compute local, structural, hydraulic, or propagated values and then order components for inspection or maintenance; the resulting ranking inherits the meaning and limitations of the underlying score (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

2.12.1 Drainage Prioritization Evidence Matrix

The following study-level comparison makes the progression from local condition assessment to structural, hybrid, risk-path, and learned hydraulic approaches explicit.

Table 2.3. Drainage prioritization evidence matrix

Study	Primary task	Representative information used	Prioritization / output	Relevance to the present problem
Iqbal et al. (2021)	Culvert blockage classification	Image-derived visual features	Local blockage class	Provides local condition information but not propagated network context
Vandaele et al. (2024)	Trash-screen blockage monitoring	Camera imagery and learned blockage features	Asset-level blockage information for maintenance/flood management	Demonstrates operational value of automated local disturbance information
Simone et al. (2022)	Vulnerability and monitoring analysis	Topology, intrinsic relevance, flow direction	Tailored node/edge centrality ranking	Shows that raw topology alone may not represent functional importance
Simone (2023)	Critical-pipe identification	Connectivity and relevance-based centrality	Pipe criticality ranking	Supports network-aware prioritization but not current disturbance propagation
Dastgir et al. (2023)	Critical-pipe analysis	Runoff paths, edge betweenness, capacity-related information	Critical-pipe ranking compared with hydrodynamic results	Demonstrates lightweight graph screening and its accuracy/complexity tradeoff
Meijer et al. (2023)	Weakest-link analysis	Storage, discharge capacity, and graph flow paths	Subsystem weakness / flood-performance screening	Shows that prioritization may combine structure with operating-capacity information

Study	Primary task	Representative information used	Prioritization / output	Relevance to the present problem
Zhang et al. (2025)	Siltation-chain risk analysis	Pipeline risk factors, centrality, edge vulnerability, propagation paths	Critical nodes and high-risk pathways	Closest drainage precedent for propagated risk-path prioritization, but with siltation-specific semantics
Zhang et al. (2024)	Hydraulic-state surrogate prediction	Graph topology, recent hydraulic states, runoff, control/boundary information	Predicted hydraulic variables rather than a direct risk ranking	Defines the learned/high-fidelity boundary of the proposed lightweight risk-index model

The matrix shows that existing studies already provide local condition signals, graph-based criticality, capacity-aware screening, propagation pathways, and richer hydraulic state modeling. These strands remain distributed across methods with different purposes and state semantics, which creates an integrative question rather than an absence of ranking methods (Iqbal et al., 2021; Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024, 2025).

2.13 Parameter Patterns in Existing Approaches

An important model-design question concerns how many parameters should be included in the generic core and its domain-specific instantiations. The literature suggests that the appropriate number cannot be fixed in advance; parameter choice follows the semantics and purpose of the model (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024, 2025).

Drainage approaches already illustrate this diversity. Standard centrality may use only graph structure, whereas tailored centrality can incorporate intrinsic node relevance and flow direction (Simone et al., 2022), (Simone, 2023). Dastgir et al. (2023) combine runoff-related path information with capacity-related information for critical-pipe analysis. Meijer et al. (2023) focus on subsystem storage and discharge capacities together with graph-based flow paths. Zhang et al. (Zhang et al., 2025) combine pipeline risk factors, multiple centrality measures, edge vulnerability, and propagation-path information. Hydraulic and learned-surrogate approaches use substantially larger sets of rainfall, boundary, geometric, control, and hydraulic-state variables (Zhang et al., 2024).

These studies show that more parameters are not automatically better, just as fewer parameters are not automatically preferable. Each added variable should represent a distinct mechanism or provide empirically useful domain information. Adding several correlated or weakly justified parameters may

reduce interpretability without improving the decision task (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Across the reviewed methods, a defensible reusable core retains only semantically distinct roles while allowing each domain to construct those roles from several observed variables where justified. For example, a drainage susceptibility mapping may combine hydraulic or physical attributes without requiring a generic propagation equation to absorb every raw parameter (Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025).

Thus, the literature supports a generic core plus domain-supported parameterization rather than a universal fixed list of physical variables. The final number of drainage-specific parameters should be determined from evidence, data availability, and validation rather than from an arbitrary target count (Simone et al., 2022; Dastgir et al., 2023; Zhang et al., 2024).

Table 2.4. Parameter patterns in existing approaches

Example approach	Representative information used	Role in model	Literature-derived implication
Tailored centrality (Simone et al., 2022; Simone, 2023)	Topology, intrinsic relevance, flow direction	Improves structural ranking	Topology can be enriched, but added variables need clear semantics
Critical-pipe graph method (Dastgir et al., 2023)	Runoff/path information and capacity-related information	Approximates criticality/flooding consequence	Shows value of separating network paths from capacity response
Weakest-link method (Meijer et al., 2023)	Storage, discharge capacity, flow paths	Screens subsystem weakness	Domain variables may be combined without redefining the generic graph concept
Siltation-chain analysis (Zhang et al., 2025)	Pipeline risk, centrality, edge vulnerability, risk path	Ranks critical nodes and paths	Multiple parameters are justified when each represents a distinct mechanism
GNN hydraulic surrogate (Zhang et al., 2024)	Hydraulic states, runoff, boundary, control variables, and graph topology	Predicts physical hydraulic states	High parameter count supports richer prediction but changes the task and data burden

2.14 Reusable Architectures and Domain-Specific Instantiation

A model can be generic in formulation without being empirically generalizable to every domain. Network science repeatedly reuses architectural ideas—nodes, directed relations, current state, transmission, response, and aggregation—while redefining their meanings for influence, epidemics, cascading infrastructure,

cybersecurity, finance, transportation, and drainage (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Faraji et al., 2026; Petrone and Latora, 2018; Simone et al., 2022).

This distinction is important because the word *generalizable* can imply demonstrated performance on unseen domains. A reusable architecture supports cross-domain instantiation only when each application can defensibly map its data and semantics to the abstract roles and validate the resulting interpretation (Kavallieratos et al., 2021; Faraji et al., 2026; Petrone and Latora, 2018).

The core propagation logic can therefore remain fixed while a domain supplies its own parameter definitions and, where justified, an additional refinement layer. Drainage, cybersecurity, transportation, and supply-chain applications require different parameter definitions, evidence, and independent validation even when they share a graph-based computational pattern (Kavallieratos et al., 2021; Zhou et al., 2023; Petrone and Latora, 2018; Dastgir et al., 2023).

This structure resembles a reusable computational backbone with application-specific adaptation, but it should not be interpreted as machine-learning transfer learning. A learned surrogate transfers fitted representations or parameters; an explicit architecture instead reuses a declared computational rule whose variables must be reinterpreted and validated in each domain (Zhang et al., 2024).

2.15 Synthesis of Research Problems

The reviewed evidence supports three major research problems.

First, local condition does not by itself represent network-context priority. Local monitoring can quantify condition at a component, but comparative prioritization in an interconnected system requires consideration of propagated network effects (Iqbal et al., 2021; Vandaele et al., 2024; Simone et al., 2022; Dastgir et al., 2023).

Second, structural importance does not by itself represent current operating response. Centrality and topology use the network as context, but drainage and broader infrastructure studies repeatedly augment structure with hydraulic relevance, capacity, storage/discharge relationships, vulnerability, or other operating information (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Buldyrev et al., 2010; Faraji et al., 2026).

Third, existing propagation states are designed for different analytical objectives. Activation, probability, infection, physical load, failure, financial loss, and hydraulic state each imply different update and aggregation rules. A bounded continuous non-probabilistic comparative risk state therefore requires an explicitly declared semantic and compatible propagation architecture (Kempe et al., 2003; Kermack and McKendrick, 1927; Sheyner et al., 2002; Buldyrev et al., 2010; Faraji et al., 2026; Zhang et al., 2024).

Together, these problems create a need for a model that can integrate local disturbance, current propagated state, and relation-specific operating response while remaining interpretable enough to explain why one node receives a higher comparative risk state than another (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

2.16 Architectural and Integrative Research Gap

The literature does not support claiming that graph propagation, recursive updating, capacity-based response, bounded state, edge weighting, or multi-source aggregation are individually novel. Each has precedent in established model families (Kempe et al., 2003; Buldyrev et al., 2010; Muñoz-González et al., 2017; Faraji et al., 2026; Petrone and Latora, 2018).

The gap is instead architectural and integrative. Within the reviewed drainage, infrastructure, cybersecurity, finance, transportation, cascade, epidemic, influence-diffusion, and graph-surrogate evidence, no clearly established lightweight directed-graph formulation was identified that combines all of the following under one explicit comparative-risk semantics: a bounded continuous non-probabilistic node state; recursive propagation of current upstream risk; explicit separation of directed reachability, optional transmission, and relation-specific susceptibility; local disturbance at receiving nodes; bounded aggregation of converging contributions; and explicit recognition that shared-path or common-cause dependence is distinct from local aggregation. This gap statement is a synthesis bounded to the literature set examined in this chapter (Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2025; Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002).

The reviewed evidence therefore establishes an architectural and integrative research gap rather than the absence of individual graph, propagation, or ranking mechanisms. A response to this gap would need to combine a clearly defined bounded state, recursive updating, relation-specific modifiers, receiving-node disturbance, convergence, and explicit dependence limits while preserving the distinction between generic computation and domain-specific evidence (Kempe et al., 2003; Buldyrev et al., 2010; Sheyner et al., 2002; Dastgir et al., 2023; Zhang et al., 2025).

Figure 2.1 summarizes how the three reviewed evidence streams lead to this architectural and integrative gap.

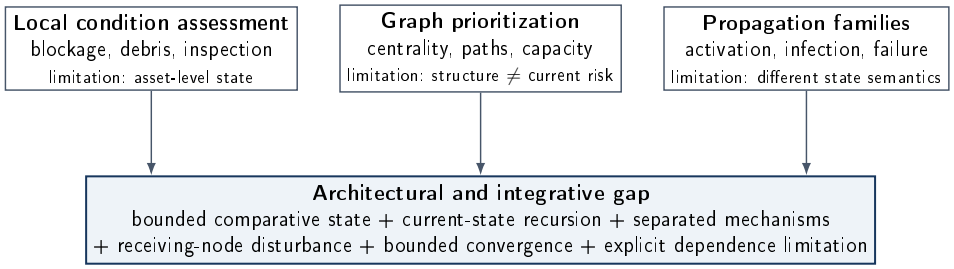


Figure 2.1. Literature-derived synthesis of existing approach families and the remaining architectural gap.

2.17 Chapter Synthesis

The literature establishes a clear progression from practical drainage prioritization to a broader network-risk modeling problem. Local assessment provides necessary component-level condition information, graph metrics provide structural context, and high-fidelity models provide domain-rich physical behavior.

Broader propagation research contributes mechanisms for recursive updating, transmission, susceptibility, aggregation, and dependence. However, these mechanisms are distributed across models with different state semantics and purposes (Kempe et al., 2003; Kermack and McKendrick, 1927; Buldyrev et al., 2010; Sheyner et al., 2002; Kavallieratos et al., 2021; Petrone and Latora, 2018).

The drainage literature also demonstrates that prioritization methods already range from local condition scoring to tailored centrality, weakest-link analysis, siltation-chain risk, hydrodynamic consequence analysis, and learned hydraulic surrogates. The research opportunity is therefore not to introduce ranking itself, but to formulate a distinct comparative-risk semantics and propagation process that can explain why one node ranks above another after both local and network context are considered (Iqbal et al., 2021; Simone et al., 2022; Dastgir et al., 2023; Meijer et al., 2023; Zhang et al., 2024).

Accordingly, the reviewed literature supports a methodological response that declares its state semantics, separates topology from relation-level response, propagates current state recursively, bounds converging contributions, and limits its claims to comparative prioritization. The specific formulation adopted by the study is presented in Chapter 3 so that this chapter remains a synthesis of prior work rather than a duplicate statement of the proposed model (Kempe et al., 2003; Buldyrev et al., 2010; Muñoz-González et al., 2017; Dastgir et al., 2023; Zhang et al., 2025).

CHAPTER III
METHODOLOGY

3.1 Research Design

The study adopts a **quantitative model-design, computational-validation, and cross-domain evaluation research design**. Its purpose is to formulate, implement, and evaluate the Graph-Based Bounded Risk Propagation Model (GB-BRPM) as a lightweight directed-graph comparative-risk propagation and prioritization model. The first evaluation layer tests internal mathematical behavior, controlled mechanism isolation, comparative prioritization, robustness, dependence limitations, and computational efficiency. The second layer examines whether the frozen architecture can be mapped coherently to drainage, software-dependency, and electrical-network data without changing its core update rule. The design does not establish universal cross-domain performance or hydraulic or electrical predictive equivalence.

The study does not use the synthetic networks to estimate universal model parameters. Instead, synthetic graph structures and bounded inputs are treated as **controlled mechanism-isolation inputs**. This permits recursive propagation, branching, convergence, reconvergence, susceptibility variation, local disturbance, and multi-source aggregation to be examined under reproducible conditions. The generic architecture remains fixed, whereas the meanings and calculations of local disturbance, susceptibility, and optional transmission belong to the selected domain instantiation and require domain-specific justification.

3.1.1 Alignment of Objectives, Methods, and Outputs

The methodology is derived directly from the objectives established in Chapter 1. The following mapping makes explicit how each objective is operationalized and what evidence or output is produced.

Table 3.1. Alignment of objectives, methods, and reported evidence.

Objective component	Methodological response	Reported evidence or output
Objective 1: Synthesize limitations and recurring mechanisms relevant to comparative network-risk prioritization	Chapter 2 literature synthesis provides the pre-methodological basis for the variables, state semantics, comparison conditions, and scope boundaries adopted in this chapter.	Literature-supported design requirements and architectural research gap

Table 3.2. Layered validation and cross-domain evaluation design.

Layer	Purpose	Data basis	Claim supported
Layer 1 — Architecture validation	Determine whether the bounded recursive formulation behaves as mathematically and computationally intended.	Controlled synthetic DAGs, reconstructed validation inputs, bounded scenario values, perturbation trials, and scalability graphs	Internal behavior, mechanism isolation, ranking response, robustness, dependence limitations, and computational properties
Layer 2 — Cross-domain evaluation	Test whether the frozen generic roles can be instantiated coherently without changing the core rule.	Drainage and SWMM scenarios, a real software dependency graph with vulnerability evidence, and real electrical topology with baseline and Component 2 phasor measurements	Behavioral and reference evidence for drainage, structural and sensitivity evidence for software, and electrical preprocessing plus controlled mechanism and sensitivity evidence

Architecture validation and cross-domain evaluation answer different questions. Controlled tests establish intended behavior; domain cases test mapping and applicability within their available evidence. A case is not described as outcome validation unless an independent reference for the relevant outcome is available. This distinction is especially important for the electrical case, whose baseline measurements verify the importer and normal-operation coverage, while the Component 2 suite tests controlled sensitivity on measured loading without supplying independent switching-event outcomes.

3.2 Graph Representation and Model Assumptions

The generic model is defined on a directed graph:

$$G = (V, E) \qquad \text{(Equation 3-1)}$$

where V is the set of entities and E is the set of directed relations. For GBBRPM v1, the evaluated graph is restricted to a **directed acyclic graph (DAG)** so that node states can be computed once in topological order from upstream to downstream. The topological sequence is a computational evaluation order

and is not interpreted as physical time.

Each node i carries a normalized continuous risk state:

$$R_i \in [0,1] \qquad \text{(Equation 3-2)}$$

where R_i is a **dimensionless normalized risk index rather than a probability**. Accordingly, a value such as $R_i = 0.70$ indicates a relatively high modeled risk under the adopted normalization and does not represent a 70% probability of failure.

For each directed relation $i \to j$, the incoming risk contribution is defined as:

$$Q_{ij} = S_{ij}\tau_{ij}R_i \qquad \text{(Equation 3-3)}$$

where $S_{ij} \in [0,1]$ is relation-specific operating susceptibility, $\tau_{ij} \in [0,1]$ is an optional transmission or influence factor, and R_i is the current propagated upstream risk.

The receiving-node risk is computed as:

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}) \qquad \text{(Equation 3-4)}$$

where $B_j \in [0,1]$ is the local disturbance severity and $P(j)$ is the predecessor set of node j.

3.3 Interpretation of Model Components

GBBRPM separates five conceptual roles. **Topology or reachability** determines where influence can travel. **Transmission or influence** represents how strongly influence transfers along an existing relation. **Susceptibility** represents the relation-specific or receiving-context response under the evaluated condition. **Local disturbance** represents risk introduced directly at a node. **Dependence** refers to shared-origin, shared-path, or common-cause effects and is not assumed to be resolved by the local aggregation rule.

The multiplicative-complement operator is used as a **bounded saturating aggregation function**. Its algebraic form is related to noisy-OR formulations, but it is not assigned probabilistic semantics in GBBRPM. Its suitability is evaluated through required mathematical behavior: boundedness, monotonicity, zero-contribution behavior, saturation, and simultaneous representation of inherited and local contributions.

Figure 3.1 provides the generic node example requested during consultation. It deliberately uses abstract nodes rather than a drainage, software, or electrical topology. The figure shows how a local input enters a node, how the current upstream state is modified by relation-specific susceptibility and optional transmission, how multiple relations converge at a receiver, and how the final computed states become the basis for comparative ranking. Domain cases later replace these abstract labels with evidence-supported meanings without changing the core update rule.

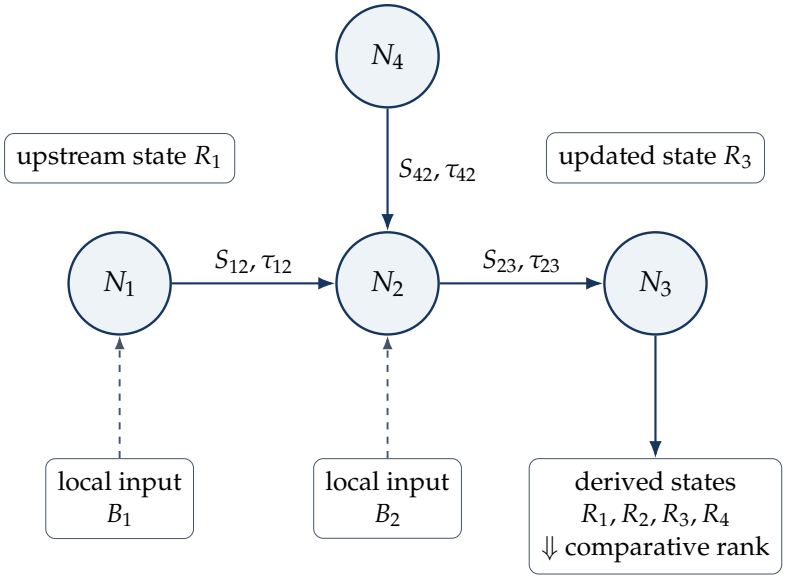


Figure 3.1. Generic directed-node example showing local inputs, relation-level modifiers, recursive state propagation, convergence at N_2, and ranking from the computed node states.

For the v1 baseline, neutral transmission is used:

$$\tau_{ij} = 1 \qquad \textit{(Equation 3-5)}$$

This avoids introducing unsupported heterogeneous transfer coefficients. Under this setting, transmission remains an explicit architectural interface but does not numerically differentiate edges. Consequently, v1 does not claim to validate calibrated heterogeneous transmission effects. Uniform non-neutral values are varied only in explicitly labeled sensitivity experiments in the software and Component 2 cases; those sweeps test attenuation behavior and are not empirical transmission calibration. The implemented baseline evaluates topology-defined reachability, susceptibility, current-state recursion, local disturbance, and bounded aggregation.

3.4 Overall Methodological and Computational Procedure

3.4.1 End-to-End Research Workflow

The complete methodology follows a traceable sequence from input preparation to scope evaluation:

1. **Select and document the input configuration.** Identify whether the run uses a controlled synthetic topology, the historical reconstructed validation inputs, the SWMM external-reference configuration, the software-dependency case, or the electrical-network case. Record the source, version or retrieval date, coverage, permitted use, and whether each field is observed, documented, derived, or assumed.

2. **Prepare and validate the graph data.** Load the node and edge tables, confirm unique identifiers and valid directed relations, check that bounded

November 14–15, 2024 coverage does not overlap the requested November 13 switching windows. Each dataset is therefore reported according to its actual provenance and evidentiary role rather than being treated as interchangeable empirical validation.

The working acquisition specification and formal request draft are maintained separately as the operational data-acquisition workspace.

The computational experiments vary the following factors while holding unrelated inputs fixed whenever possible:

- local disturbance severity B_j;
- source or disturbance location;
- relation-specific susceptibility S_{ij};
- number and placement of simultaneous sources;
- intermediate local disturbance;
- graph structure;
- shared-origin reconvergence; and
- small perturbations of bounded model inputs for robustness analysis.

The restored historical core protocol contains **136 controlled scenario runs** distributed as follows:

Table 3.5. Controlled 136-run scenario protocol.

Scenario family	Design	Runs
Source severity	Four severity levels across N1–N5	20
Drainage L/C stress	Five load-scaling levels across N1–N5 using the selected drainage susceptibility mapping	25
Disturbance location	Historically selected local-disturbance locations by network	36
Multi-source configuration	Multi-source configurations for N3 and N5	10
Intermediate disturbance	Three selected intermediate nodes $\times$ three disturbance severities across N1–N5	45
Total	**Restored controlled scenario protocol**	**136**

The severity sweep uses $B \in \{0.25, 0.50, 0.75, 1.00\}$. The drainage L/C stress sweep multiplies edge load values by $\{0.50, 0.75, 1.00, 1.25, 1.50\}$, after which susceptibility is recomputed using the selected clipped-linear mapping. Disturbance-location experiments isolate one selected local disturbance at severity 0.75. Multi-source-configuration experiments preserve the historical source

severities for the active source subset. Intermediate-disturbance experiments preserve the network's baseline local disturbances and impose the selected intermediate disturbance at severities 0.25, 0.50, and 0.75.

Non-neutral τ_{ij} values are not part of the 136-run v1 baseline protocol. They may be used in a clearly labeled controlled sensitivity sweep without being interpreted as calibrated transmission; operational interpretation would require an independently defensible domain-specific construction.

3.7 Baselines and Comparative Experiments

Three primary comparison conditions are used to determine what is added by condition-aware propagation:

1. **Local-only baseline:** $R_j = B_j$, with no propagated contribution.
2. **Uniform-susceptibility propagation:** the same recursive bounded architecture is retained, but all relations share a constant susceptibility value.
3. **Heterogeneous-susceptibility GBBRPM:** relation-specific susceptibility values are used while the baseline transmission factor remains $\tau_{ij} = 1$.

These comparators are diagnostic baselines rather than ground truth. Their purpose is to determine whether heterogeneous relation conditions materially alter downstream magnitude and ranking beyond local scoring or uniform propagation.

The comparator set is designed for **mechanism isolation**, not for claiming universal superiority over propagation models with different state semantics. Independent Cascade and Linear Threshold models represent activation processes, SIR-type models represent compartmental state transitions, centrality methods measure structural position, and hydraulic simulators represent physical system states. Their outputs are therefore not directly interchangeable with the bounded continuous non-probabilistic R_i state. Those approaches are compared conceptually in Chapter 2 to establish the research gap and design choices. The experimental baselines in this chapter instead hold most of the GBBRPM architecture constant while selectively removing propagation or heterogeneous susceptibility, allowing the added effect of each selected mechanism on magnitude and ranking to be interpreted directly.

3.8 Mathematical and Behavioral Validation

The first validation layer tested whether the model behaved consistently with its declared state semantics and mathematical design.

Boundedness. All computed node states must satisfy:

$$0 \leq R_j \leq 1 \qquad \text{(Equation 3-9)}$$

Monotonicity. Increasing a genuine risk contribution while holding other inputs

fixed should not decrease the resulting node risk. For $x_k = Q_{kj}$,

$$\frac{\partial R_j}{\partial x_k} = (1 - B_j) \prod_{i \neq k} (1 - x_i) \geq 0 \qquad \textit{(Equation 3-10)}$$

and

$$\frac{\partial R_j}{\partial B_j} = \prod_i (1 - x_i) \geq 0 \qquad \textit{(Equation 3-11)}$$

which establishes local monotonicity of the aggregation rule over the declared bounded input space.

Boundary tests additionally verified zero-contribution behavior, $S_{ij} = 0$ edge gating, and the expected limiting behavior when $B_j = 1$.

3.9 Sensitivity and Scenario Analysis

Sensitivity analysis evaluated how risk magnitude, ranking, and propagation reach changed when model inputs were altered. Source severity was varied across bounded levels, and susceptibility was varied through controlled scaling or direct bounded test values. Multi-source configurations and intermediate disturbances were modified to observe how multi-source convergence and locally introduced risk affected downstream modeled risk.

The analysis recorded node-level risk values together with summary indicators such as maximum risk, mean risk, affected-node count under explicitly declared reporting thresholds, propagation depth, top-ranked nodes, top-k overlap, and designated outlet risk where applicable. Reporting thresholds are analytical cutoffs only and are not interpreted as physical failure probabilities.

3.10 Robustness and Ranking Stability

Input robustness was evaluated by perturbing bounded model parameters around a reference condition. Three perturbation magnitudes were used: **$\pm$5%, $\pm$10%, and $\pm$20%**. For the historical validation protocol, each perturbation level was evaluated across **600 trials** using a fixed random seed of **41** to support reproducibility.

Primary ranking metrics include **Spearman rank correlation** between the reference and perturbed rankings and **top-3 Jaccard similarity** for the highest-ranked nodes. The analysis may also record top-3 appearance frequency for individual nodes to identify whether high-priority nodes remain consistently represented under perturbation.

This analysis distinguishes a model that produces stable prioritization under modest uncertainty from one whose rankings change substantially under small input variation. Ranking instability, when observed, is reported as a model characteristic rather than automatically treated as failure.

3.11 Reconvergence and Dependence Diagnostics

Reconverging networks were used to evaluate shared-origin effects. When one upstream disturbance splits into multiple paths and later reconverges, the local bounded aggregator may treat those incoming path contributions as separate even though they share a common origin. This produces a possible overlap-inflation effect.

Accordingly, reconvergence was evaluated separately from ordinary multi-source aggregation. The methodology adopts the principle:

$$\text{bounded aggregation} \neq \text{dependence correction} \qquad \textit{(Equation 3-12)}$$

The v1 model does not retain complete source ancestry and therefore does not claim to resolve common-cause or shared-path dependence.

3.12 Computational Scalability

Computational cost was assessed on increasingly large sparse DAGs. Runtime was measured while graph size and edge count increased. Because the evaluator performs a topological traversal and processes each node and edge once under the implemented DAG formulation, the expected computational behavior is consistent with:

$$O(|V| + |E|) \qquad \textit{(Equation 3-13)}$$

for a one-pass implementation. Runtime measurements were reported as empirical implementation results rather than as proof of universal performance across all graph representations or software environments.

3.13 Reproducibility and Software Implementation

The model is implemented as a deterministic Python workflow using structured node and edge input tables. Experiment scripts generate scenario outputs, comparator results, robustness summaries, property-test results, and scalability measurements. Randomized experiments use fixed seeds so that trial generation can be reproduced.

The repository preserves the model evaluator, restored historical scenario runner, comparison routines, robustness tests, property checks, input tables, and result-export workflow under version control. The experiment runner supports explicit data-source selection so that the historical reconstructed validation inputs can be evaluated separately from an alternate recovered historical candidate dataset.

The primary thesis-reproduction command uses the historical source configuration. Generated result files are traceable to the corresponding input source, code revision, scenario protocol, trial count, and random seed.

Data-provenance status. The exact lost original raw edge-by-edge input table has not been independently recovered. However, the earlier

`validation_reconstructed_edges` and `validation_reconstructed_nodes` inputs have been recovered and computationally verified to reproduce the preserved historical baseline and comparator behavior and to restore the 136-scenario protocol structure and run count. They are therefore treated as **historical reconstructed validation inputs**, not as newly invented fixtures and not as the lost original raw dataset. A separate recovered historical candidate dataset is retained for provenance comparison because it produces a different numerical configuration.

3.14 Cross-Domain Instantiation and Evaluation

The frozen GBBRPM core was evaluated in three domains. The cases were selected to test complementary claims rather than to imply equal levels of empirical validation. Across all cases, nodes, directed relations, local evidence, relation-specific modifiers, and output semantics were documented before interpretation.

3.14.1 Drainage Case

The drainage case used the five controlled DAGs N1–N5, the restored 136-run scenario protocol, robustness perturbations, and 12 SWMM blockage scenarios. The controlled experiments evaluated bounded behavior, recursive propagation, ranking sensitivity, robustness, and dependence limitations. The SWMM comparison used mean Spearman rank correlation, top-20% Jaccard overlap, and the share of hydraulic worsening outside the downstream-only GBBRPM scope. SWMM outputs were treated as a hydraulic reference, not as observed field ground truth.

3.14.2 Software-Dependency Case

The software case represented packages as nodes and dependency-to- dependent relations as directed edges. The analyzed Express 4.18.2 topology contained 71 packages and 128 directed dependency edges. Evidence-backed affected versions supplied bounded local disturbance levels based on advisory severity. A transmission-coefficient sweep used $\tau \in \{0, 0.25, 0.50, 0.75, 1.00\}$ while retaining the same core aggregation rule. Express root risk, maximum network risk, rank, Spearman agreement with the $\tau = 1$ ranking, and top-20% Jaccard overlap were reported. This case tests real-topology structural applicability and sensitivity; it does not claim prediction of future exploitation.

3.14.3 Electrical-Network Case

The electrical case had two complementary stages. First, the full electrical importer represented bus records as nodes and active connections as directed edges. The imported topology contained 187 bus records and 177 active connections. Baseline magnitude archives were summarized in chunks, with empty CSV members recorded rather than causing the run to fail. Channel mapping, populated and empty files, inferred one-second timestamp gaps, meter-level voltage coverage, and usable voltage channels were reported.

Component selection rationale. Component 2 was selected purposively

before interpreting scenario results because it satisfied the requirements of the intended electrical mechanism study: an authorized topology extract; synchronized phasors at a traceable source boundary and multiple terminal loads; a nontrivial branching structure with internal and terminal nodes; and an observable directed subgraph compatible with the v1 DAG evaluator. These conditions permitted edge loading to be derived by documented downstream-meter aggregation rather than invented for unmetered branches. The selection was therefore based on data availability, observability, traceability, and model compatibility, not on obtaining a favorable risk ranking. Component 2 is treated as a fit-for-purpose case and is not claimed to be statistically representative of the entire SoCal 28-Bus system or of distribution grids generally.

Second, an authorized extract of Component 2 from the Caltech SoCal 28-Bus digital-twin dataset was used for a controlled real-topology, measured-loading instantiation (Xie et al., 2025). Zero-impedance-connected buses were consolidated into model nodes. The complete component contained 13 nodes and 12 directed elements; one unobserved element (`line_407`) was excluded, leaving an observable arborescence of 12 nodes and 11 edges. Direction followed the retained topology `fbus`-to-`tbus` orientation.

Twenty-four synchronized phasor snapshots beginning at November 14, 2024 07:00 were summarized. The source meter was `egauge_21`; five terminal-load meters supplied downstream real and reactive power. `egauge_19` was excluded because it measured a neighboring feeder, and `egauge_15` lacked complete usable phasors. The mean source power was 985.957 kW and the five metered loads summed to 976.828 kW. The 9.129 kW real-power residual and 135.398 kvar reactive-power residual were reported separately and were not allocated to unmetered nodes.

For each observable edge, loading was the mean downstream apparent power L_{ij} derived from the phasors. Transformer capacity was the minimum declared nameplate kVA when multiple values existed. Line capacity was computed as three times phase-to-ground nominal voltage times the declared current rating. Electrical susceptibility was then instantiated as $S_{ij} = \min(1, L_{ij}/C_{ij})$. Local disturbance was initialized to $B_i = 0$, and transmission was initialized to $\tau_{ij} = 1$, except when explicitly varied in a controlled scenario. The declared `Mains_Power` register was used only as a cross-check because its metadata unit and raw numerical scale were inconsistent; phasor-derived power remained the model input.

The controlled suite contained 43 deterministic runs: one zero-disturbance reference; four root-severity levels; disturbances at each of the 12 observable nodes; five uniform measured-load scales; five uniform transmission levels; 15 pairs of internal non-root sources; and one all-internal-source case. Root-severity, location, load-scale, and transmission experiments used a declared disturbance rather than an observed fault label. The Component 2 graph is an arborescence, so nested multi-source pairs test bounded local-plus-incoming aggregation and parallel pairs test concurrent branch coverage; they do not test downstream reconvergence. Reported outcomes included maximum and mean risk, risk sum, nodes above the 0.05 threshold, top-20% nodes with ties, and terminal-node risk.

The available baseline measurements cover normal operation, whereas the requested switching-event groups occurred outside the available aligned

window. Consequently, the Component 2 suite is a mechanism and sensitivity experiment on measured loading and real topology, not event-response prediction or independent outcome validation.

3.14.4 Evidence Separation and Reproducibility

The drainage, software, and electrical cases support different claims. Behavioral and external-reference evidence in drainage is not equated with software structural applicability or controlled electrical mechanism evidence. Core cross-domain paper outputs are produced from versioned inputs by `scripts/generate_paper_outputs.py`. The Component 2 scenarios and figures are produced separately by `scripts/run_caltech_component2_experiments.py` from the adapter-generated model tables. The associated manifests record input hashes, output files, scenario policies, and the interpretation boundary used in reporting.

3.15 Methodological Scope

The architecture-validation layer evaluates internal behavior, comparative-prioritization output, robustness, dependence limitations, and computational properties under controlled directed-graph conditions. The cross-domain layer evaluates instantiability within the three selected cases while preserving their distinct evidence roles.

The methodology does not establish calibrated failure probabilities, universal physical parameter mappings, universal cross-domain performance, shared-path dependence correction, bidirectional or cyclic feedback behavior, hydraulic predictive equivalence, or electrical-event prediction. Dependence-aware aggregation, cyclic or feedback behavior, out-of-sample domain studies, and event-aligned outcome validation require separate evidence or future model extensions.

CHAPTER IV
RESULTS AND DISCUSSION

4.1 Overview of Validation Results

The restored GBBRPM v1 validation pipeline was executed using the historical reconstructed validation inputs and the frozen model formulation described in Chapter 3. The controlled experiment suite produced 136 scenario runs, supplemented by comparator analysis, robustness trials, mathematical property tests, scalability measurements, reconvergence diagnostics, a SWMM external-reference comparison, and software and electrical cross-domain cases, including a separate 43-run Component 2 controlled electrical suite.

The results are interpreted according to the evidence role of each case. They are not interpreted as calibrated failure probabilities, proof that GBBRPM reproduces full hydraulic behavior, prediction of software exploitation, or completed electrical event-response validation.

Table 4.1. Evidence roles of the three evaluated domains.

Domain	Instantiated data	Evidentiary role	Status
Drainage	5 synthetic DAGs; 12 SWMM scenarios	Behavioral tests and hydraulic-reference comparison	Completed
Software	71 packages; 128 dependency edges; 7 vulnerable versions	Real-topology structural applicability and sensitivity	Completed
Electrical	187-bus baseline; 12-node observable Component 2 model; 43 controlled runs	Import verification plus real-topology, measured-loading mechanism and sensitivity tests	Controlled study complete; event validation pending

The three cases test different claims; structural applicability and controlled sensitivity are not labeled as independent outcome-prediction validation.

4.2 Baseline Behavior Across N1–N5

The baseline experiments evaluate the five synthetic DAGs under their reference local disturbances and heterogeneous edge susceptibilities. N1 isolates sequential propagation, N2 exposes branch-dependent differences, N3 demonstrates bounded multi-source convergence, N4 stresses reconvergence from a shared upstream origin, and N5 combines multiple sources, branching, convergence, heterogeneous susceptibility, and an intermediate local disturbance.

Table 4.2. Synthetic drainage-network evaluation set.

Network	Nodes	Edges	Sources	Mean S	Primary purpose
N1	6	5	1	0.560	Attenuation and propagation depth
N2	8	7	1	0.521	Branch divergence and path sensitivity
N3	8	7	2	0.621	Independent multi-source aggregation
N4	7	7	1	0.636	Reconvergence / overlap stress
N5	20	27	3	0.619	Main heterogeneous validation network

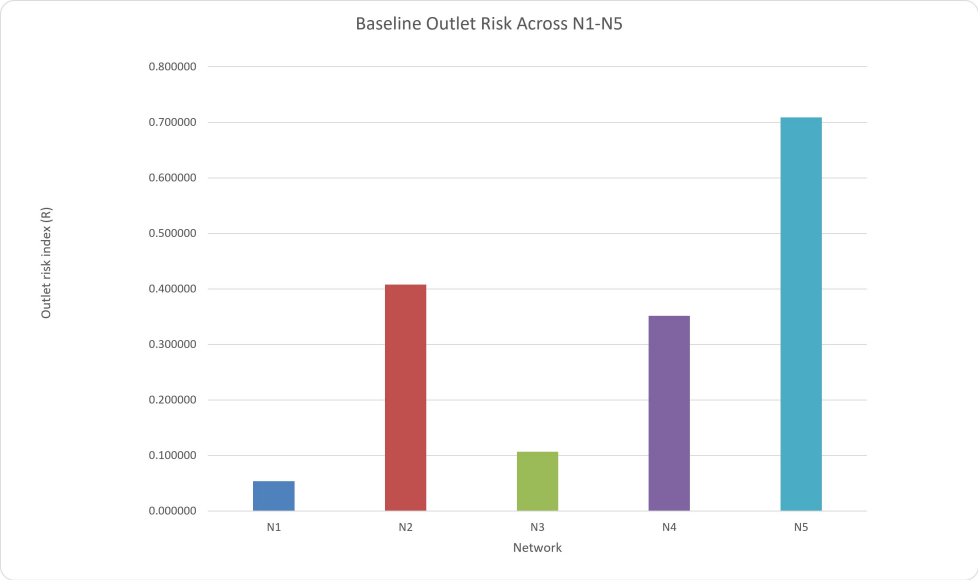


Figure 4.1. Baseline outlet risk across the five controlled GBBRPM networks under the historical validation configuration.

For the historical N5 baseline, the designated outlet risk is

$$R_{N20} = 0.7086404493. \qquad \textit{(Equation 4-1)}$$

The highest-ranked internal nodes are ordered

$$N13 > N19 > N16. \qquad \textit{(Equation 4-2)}$$

This ranking shows that the model does not simply prioritize source nodes or the final outlet. Accumulated upstream state, susceptibility, convergence structure, and local disturbance can elevate intermediate nodes.

Table 4.3. Baseline summary across the controlled network set.

Network	Max. risk	Mean risk	Outlet	Outlet risk	Top-three nodes
N1	0.800000	0.249493	F	0.053760	A > B > C
N2	0.800000	0.465250	H	0.408000	A > C > G
N3	0.700000	0.420205	H	0.106814	A > C > B
N4	0.800000	0.579719	G	0.351372	A > D > B
N5	0.816448	0.632660	N20	0.708640	N13 > N19 > N16

The networks do not exhibit a uniform propagation pattern. N1 shows strong attenuation along a chain, while N2 retains comparatively high downstream risk because several high-susceptibility branches feed later nodes. N3 demonstrates bounded convergence from two local sources, N4 produces stronger downstream amplification under reconvergence, and N5 produces a high-risk intermediate ranking rather than prioritizing the outlet alone. These patterns are consistent with the intended use of the networks as mechanism-isolation cases rather than interchangeable test graphs.

4.3 Controlled Scenario Analysis

The restored historical suite contains 136 runs distributed across five scenario families.

Table 4.4. Controlled scenario families and run counts.

Scenario family	Runs	Purpose
Source severity	20	Increasing local disturbance
L/C stress	25	Increasing operating susceptibility
Disturbance location	36	Sensitivity to disturbance placement
Multi-source configuration	10	Simultaneous source disturbances
Intermediate disturbance	45	Disturbance within propagation paths
Total	**136**	Restored controlled protocol

The archived scenario-sweep file was used to recover the historical scenario structure and parameter selections, but its stored numerical values belong to a different recovered dataset generation. Consequently, the quantitative results below were recomputed using the validated historical reconstructed node and edge inputs rather than copied from the mixed-generation archive.

4.3.1 Source-Severity Response

The source-severity sweep varies the selected source disturbance over 0.25, 0.50, 0.75, and 1.00 while preserving the remaining baseline local disturbances. All five networks show monotone outlet response.

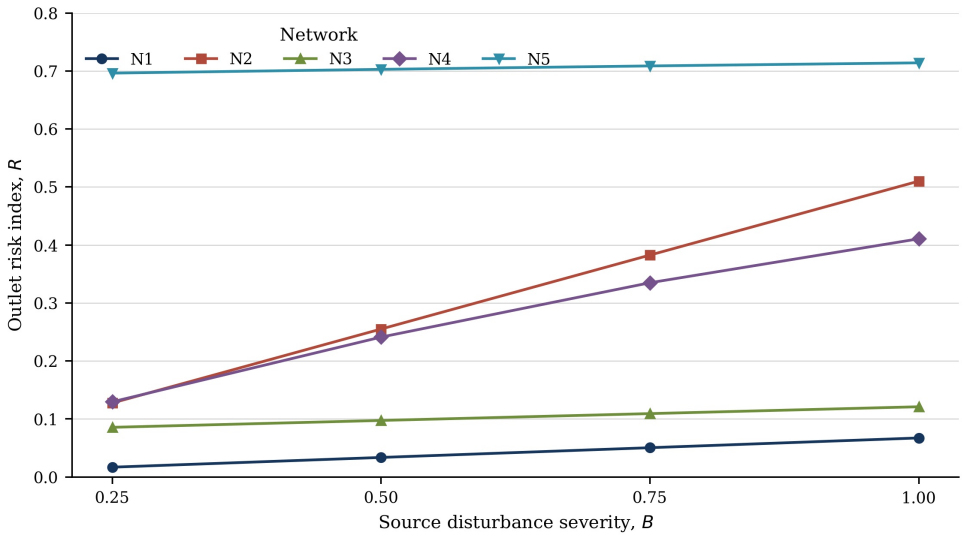


Figure 4.2. Outlet-risk response to increasing source disturbance severity across N1–N5.

Table 4.5. Source-severity response at the lowest and highest tested levels.

Network	Outlet at 0.25	Outlet at 1.00	Change	Top-three ordering, 0.25 → 1.00
N1	0.016800	0.067200	+0.050400	A > B > C → A > B > C
N2	0.127500	0.510000	+0.382500	A > C > G → A > C > G
N3	0.085549	0.120991	+0.035443	B > C > E → A > C > E
N4	0.129511	0.410550	+0.281039	A > D > F → A > B > D
N5	0.696268	0.713972	+0.017704	N13 > N19 > N10 → N1 > N13 > N5

N1 and N2 preserve their top-three ordering, whereas N3, N4, and N5 exhibit ranking changes. N5 shows only a small outlet increase because the varied N1 disturbance is embedded in a network containing other active disturbances. Thus, stronger local disturbance is directionally monotone, while ranking response depends on the surrounding network state.

4.3.2 L/C Stress Response

The stress sweep scales all edge loads by 0.50, 0.75, 1.00, 1.25, and 1.50, after which susceptibility is recomputed as $S = \min(1, L/C)$. The outlet response is strongly nonlinear in the more convergent networks.

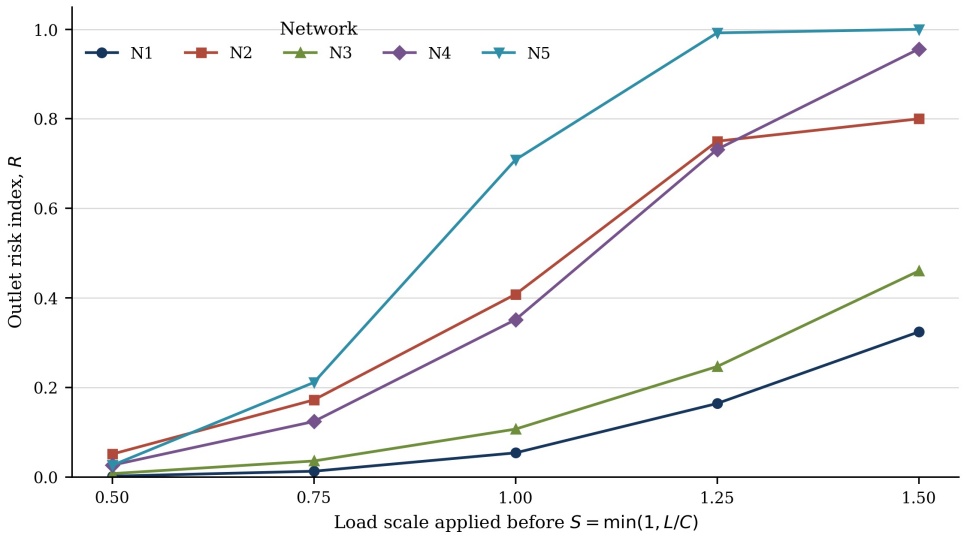


Figure 4.3. Outlet-risk response to increasing L/C stress under bounded susceptibility.

Table 4.6. Outlet-risk response across the L/C stress range.

Network	Outlet at $\times 0.50$	Outlet at $\times 1.00$	Outlet at $\times 1.50$	Top-three ordering, $\times 0.50 \rightarrow \times 1.50$
N1	0.001680	0.053760	0.324000	A > B > C → A > B > C
N2	0.051000	0.408000	0.800000	A > C > B → A > C > G
N3	0.007402	0.106814	0.460077	A > B > C → C > E > G
N4	0.026261	0.351372	0.956000	A > B > C → D > E > F
N5	0.025978	0.708640	0.999998	N1 > N3 > N2 → N19 > N20 > N18

The strongest changes occur in N4 and N5. These values demonstrate the compounding effect of increasing susceptibility along multiple recursive paths and are not hydraulic failure probabilities. Saturation follows from the clipped-linear rule: once an edge reaches $S = 1$, further increases in L/C cannot increase its susceptibility.

4.3.3 Disturbance-Location Sensitivity

The location sweep assigns the same local severity, $B = 0.75$, to historically selected locations while clearing other local disturbances. This isolates the effect of disturbance placement.

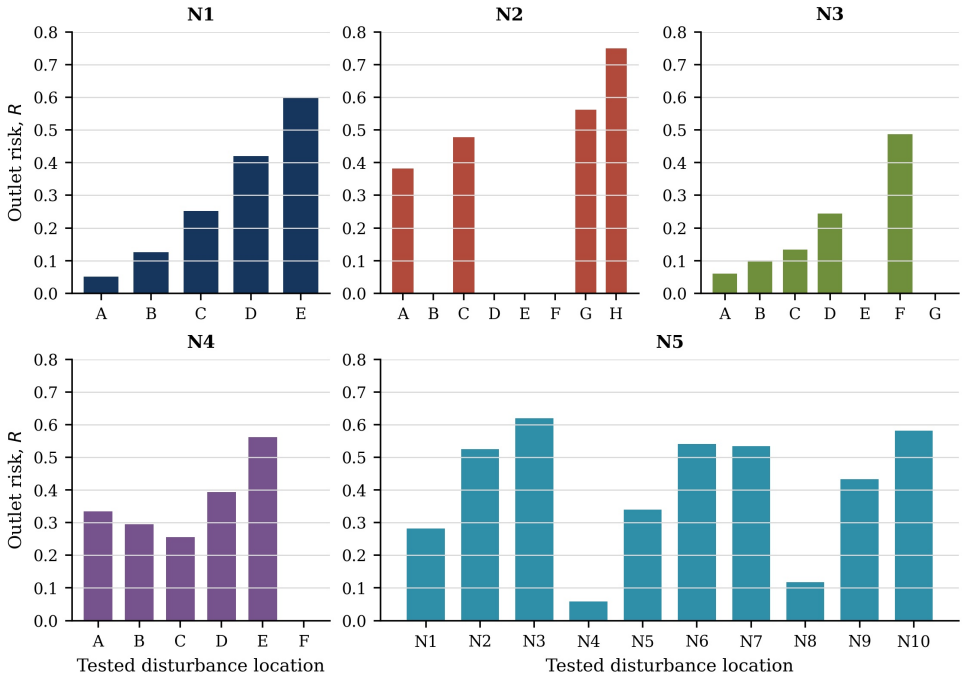


Figure 4.4. Sensitivity of outlet risk to disturbance location at fixed local severity.

Table 4.7. Range of outlet risk across tested disturbance locations, with tied extrema retained.

Network	Lowest location	Lowest risk	Highest location	Highest risk	Range
N1	A	0.050400	E	0.600000	0.549600
N2	B/D/E/F	0.000000	H	0.750000	0.750000
N3	E/G	0.000000	F	0.487500	0.487500
N4	F	0.000000	E	0.562500	0.562500
N5	N4	0.058820	N3	0.619410	0.560590

The large within-network ranges show that equal local severity does not imply equal downstream modeled risk. The result directly supports the role of topology and downstream connectivity in prioritization: local severity alone is insufficient to characterize network-level comparative risk.

4.3.4 Multi-Source Configurations

The N3 and N5 experiments activate different subsets of the historical source disturbances while preserving each source's baseline severity.

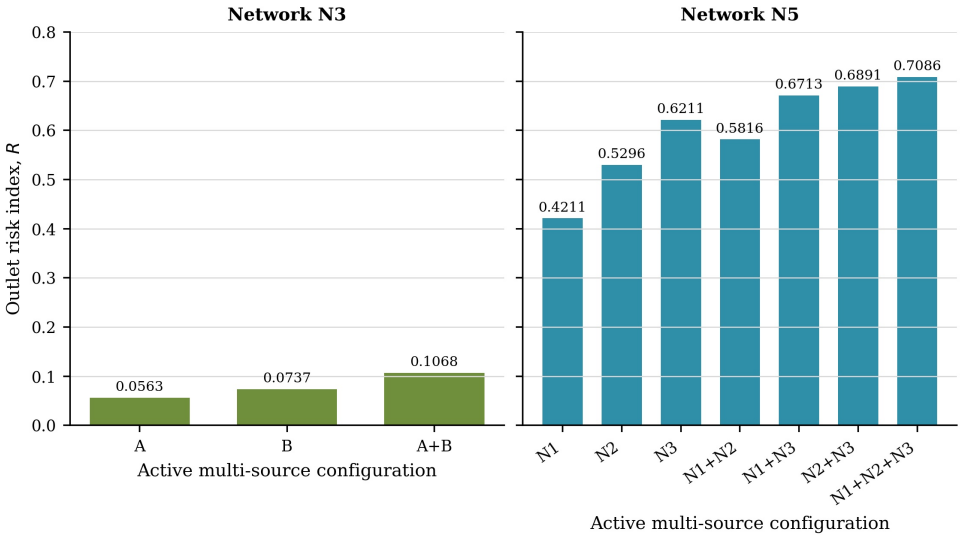


Figure 4.5. Outlet-risk response under single- and multi-source configurations in N3 and N5.

Table 4.8. Multi-source configuration results for N3 and N5.

Network	Active sources	Mean risk	Outlet risk	Top-three nodes
N3	A	0.226641	0.056306	A > C > E
N3	B	0.250959	0.073734	B > C > E
N3	A+B	0.420205	0.106814	A > C > B
N5	N1	0.310047	0.421144	N1 > N5 > N8
N5	N2	0.330667	0.529607	N19 > N16 > N13
N5	N3	0.363189	0.621147	N10 > N19 > N3
N5	N1+N2	0.475306	0.581552	N1 > N5 > N16
N5	N1+N3	0.539799	0.671334	N1 > N19 > N13
N5	N2+N3	0.503194	0.689122	N13 > N19 > N10
N5	N1+N2+N3	0.632660	0.708640	N13 > N19 > N16

Activating both N3 sources raises outlet risk above either single-source case. In N5, the single-source cases produce materially different outlet risks, and the full configuration reaches 0.7086. Pairwise combinations also alter the highest-ranked nodes. The bounded aggregator permits the contributions to coexist without exceeding 1, but does not establish probabilistic independence.

4.3.5 Intermediate Disturbance

The intermediate-disturbance protocol evaluates three selected nodes in each network at severities 0.25, 0.50, and 0.75, producing 45 runs. Baseline local disturbances are retained, so the inserted disturbance is evaluated together with inherited risk.

Table 4.9. Largest outlet-risk response in the intermediate-disturbance tests.

Network	Baseline outlet	Largest outlet	Scenario	Increase
N1	0.053760	0.613440	E=0.75	+0.559680
N2	0.408000	0.852000	H=0.75	+0.444000
N3	0.106814	0.160766	C=0.75	+0.053952
N4	0.351372	0.481593	D=0.75	+0.130221
N5	0.708640	0.852160	N19=0.75	+0.143520

The effect is highly location-dependent. N1 exhibits a large change when a severe disturbance is placed near the downstream end, while N3 shows a more modest increase. N5 reaches approximately 0.8522 when N19 is assigned $B = 0.75$. These results illustrate how a local term can coexist with recursively inherited contributions at an already affected node.

4.4 Comparator Analysis

Comparator experiments test whether heterogeneous condition-aware propagation produces ranking behavior different from simpler alternatives.

Table 4.10. N5 comparator agreement relative to GBBRPM.

Comparator	Spearman ρ	Top-three Jaccard
Local-only	0.108627	0.0
Uniform $S = 0.25$	-0.042105	0.0
Uniform $S = 0.50$	0.150376	0.0
Uniform $S = 0.75$	0.696241	0.5

Low agreement with the local-only baseline indicates that propagated network context substantially changes prioritization. Agreement also varies across uniform-susceptibility comparators, showing that ranking depends on the heterogeneous edge-condition pattern rather than only on the recursive architecture. The relatively higher agreement at uniform $S = 0.75$ is specific to this N5 configuration and is not evidence that this constant is universally preferable.

4.5 Robustness and Ranking Stability

Robustness was evaluated at perturbation magnitudes of $\pm 5\%$, $\pm 10\%$, and $\pm 20\%$, with 600 trials per level and fixed seed 41.

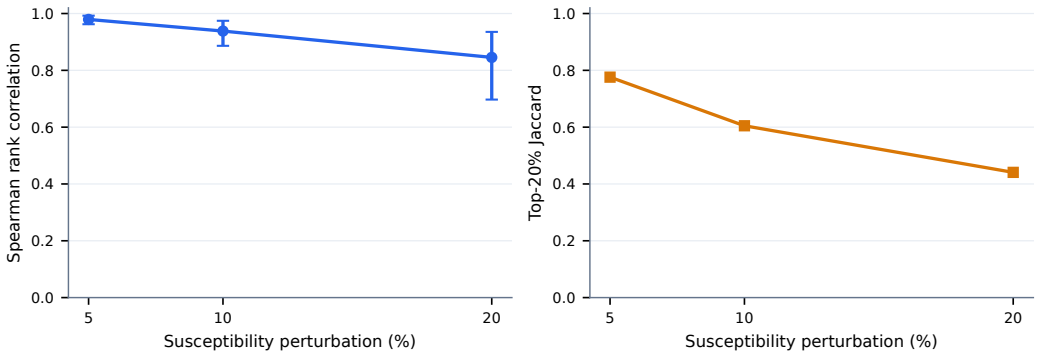


Figure 4.6. Drainage ranking stability under ±5%, ±10%, and ±20% susceptibility perturbations. Error bars show the reproduced Spearman distribution; the second panel reports top-20% Jaccard overlap.

Table 4.11. Reproduced ranking-robustness summary across 600 trials.

Perturbation	Mean Spearman	5th percentile	95th percentile	Mean top-three Jaccard
±5%	0.979190	0.962331	0.992481	0.744500
±10%	0.938267	0.886289	0.974436	0.546667
±20%	0.845552	0.697067	0.935338	0.388833

The results show high stability under small perturbations and progressively lower stability as uncertainty increases. Mean Spearman correlation remains high even at ±20%, while priority-set overlap declines more strongly. Table 4.11 reports top-three overlap, whereas Figure 4.6 uses top-20% overlap. The overall ordering may therefore remain broadly similar even when membership among the highest-priority nodes changes. Accordingly, GBBRPM is comparatively stable for broad prioritization under modest input uncertainty, while exact top-ranked membership becomes more sensitive at larger perturbation levels.

4.6 Mathematical Property Tests

Randomized property testing was executed across 600 trials. All tested cases passed the declared baseline properties.

Table 4.12. Randomized mathematical property-test results.

Property	Passes	Total
Boundedness	600	600
Monotonicity in local disturbance	600	600
$B = 1 \Rightarrow R = 1$	600	600
$S = 0$ gates the edge contribution	600	600

These randomized checks complement the analytical results in Chapter 3.

They do not prove correctness for every possible implementation or extension, but provide computational evidence that the frozen v1 implementation behaves consistently with its bounded semantics over the tested cases.

4.7 Reconvergence and Dependence Diagnostic

When one disturbance splits across paths that later reconverge, the local bounded aggregator can treat the incoming terms as distinct even though they share an origin. The executable diagnostic compares ordinary GBBRPM with an ancestry-aware path-pruned counterfactual. Independent source lineages retain multiplicative-complement aggregation; overlapping lineages are grouped and only the largest incoming contribution in each group is retained.

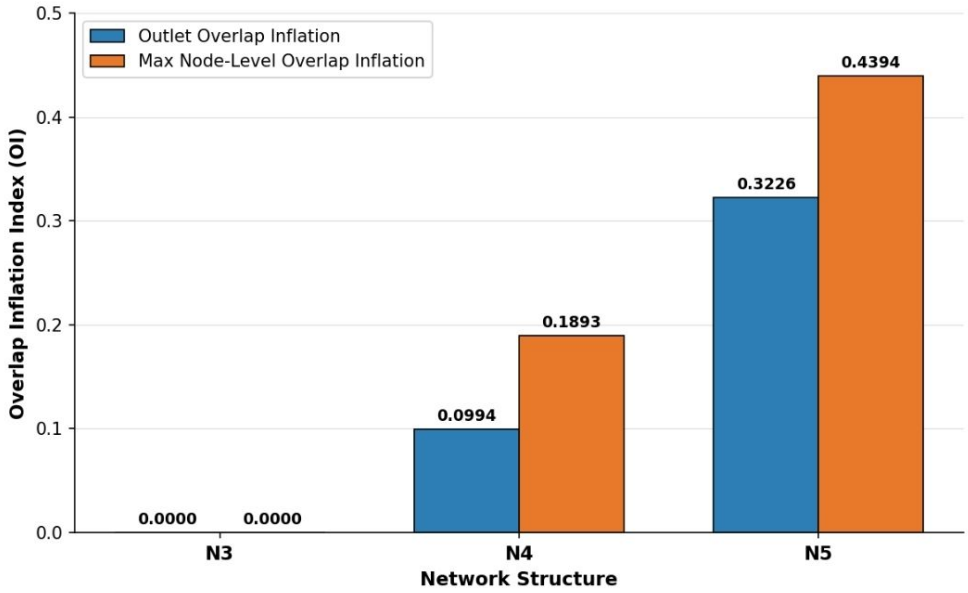


Figure 4.7. Overlap-inflation diagnostic for converging and reconverging networks.

Table 4.13. Regenerated reconvergence and overlap-inflation diagnostic.

Network	Maximum OI	Mean OI	Outlet OI	Outlet GBBRPM	Outlet path-pruned
N3	0.000000	0.000000	0.000000	0.106814	0.106814
N4	0.189280	0.083148	0.099372	0.351372	0.252000
N5	0.439353	0.149633	0.322637	0.708640	0.386003

N3 has no overlap inflation because its convergence is driven by independent source branches. N4 shows moderate inflation, and N5 shows the largest effect. The path-pruned result is not ground truth or a replacement formulation; it is a diagnostic counterfactual revealing where shared ancestry may be counted through multiple paths. The values are generated by `reconvergence.py` and exported in `results/reconvergence_summary.csv` and `results/reconvergence_node_diagnostic.csv`.

4.8 Computational Scalability

The evaluator was timed on sparse DAGs with approximately three edges per node. For each size, five warm-up evaluations preceded 30 timed repetitions. Because scheduling delays can distort the arithmetic mean, the median and interquartile range (IQR) are the primary statistics.

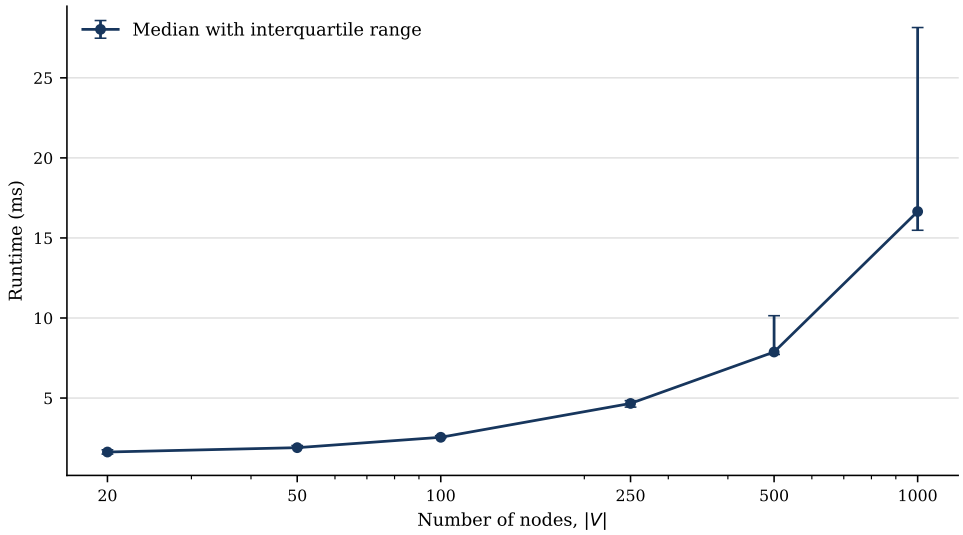


Figure 4.8. Median runtime and interquartile range of the one-pass GBBRPM evaluator on sparse DAGs.

Table 4.14. Scalability measurements after five warm-up runs and 30 repetitions.

Nodes	Edges	Median (ms)	IQR (ms)	Mean (ms)
20	60	1.628378	1.499549–1.767802	2.616805
50	150	1.900651	1.880617–2.032984	1.971181
100	300	2.548793	2.520786–2.649607	2.623430
250	750	4.663104	4.434086–4.838490	4.697808
500	1500	7.876152	7.722431–10.143945	10.103489
1000	3000	16.650025	15.479139–28.142643	24.144717

Median runtime increases monotonically from approximately 1.63 ms at 20 nodes to 16.65 ms at 1000 nodes. Differences between median and mean at several sizes reflect isolated timing outliers rather than changes in the deterministic result. The observed growth is consistent with the expected one-pass $O(|V| + |E|)$ implementation over sparse DAGs, but remains machine- and load-dependent. The complete timing distribution is retained in `results/scalability.csv`.

4.9 SWMM External-Reference Comparison

A separate SWMM workflow compared GBBRPM downstream-risk patterns with hydraulic consequences under 12 controlled conduit-blockage scenarios.

For each case, conduit area loss a is represented in GBBRPM as reduced effective capacity,

$$C'_{ij} = (1 - a)C_{ij}, \qquad \textit{(Equation 4-3)}$$

with reference load held fixed and susceptibility recomputed. Positive GBBRPM risk change and positive SWMM maximum-depth or flooding-volume change are calculated relative to their respective baselines. Spearman correlation is evaluated over the union of nodes with positive change in either workflow; top-20% Jaccard compares their highest-priority positive-change sets. Outside-scope share measures positive hydraulic worsening outside the blocked edge's target-and-descendant region.

Table 4.15. GBBRPM and SWMM ranking comparison across 12 blockage scenarios.

Hydraulic outcome	Mean Spearman	Top-20% Jaccard	Outside-scope share
Maximum depth	-0.416	0.119	71.2%
Flooding	-0.485	0.000	40.5%

Negative rank correlations and low priority-set overlap are reported without reinterpreting the modeled risk index as a hydraulic predictor.

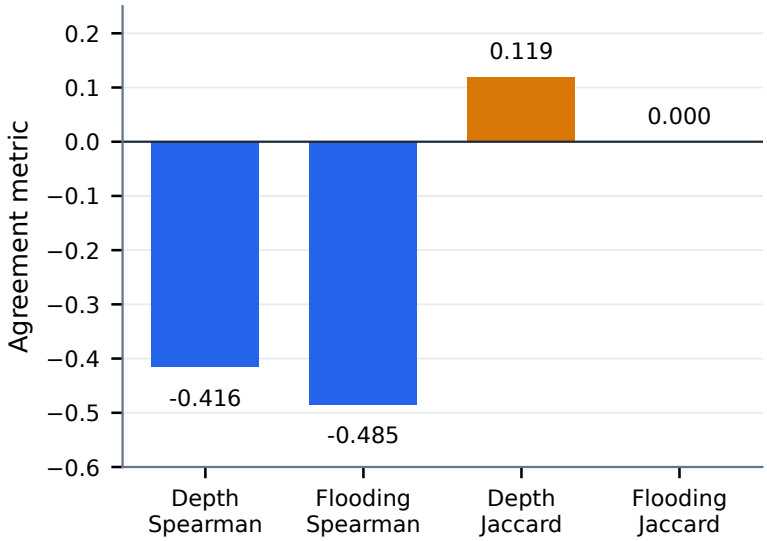


Figure 4.9. Agreement between GBBRPM priority changes and SWMM hydraulic outcomes across the 12 blockage scenarios.

The negative rank correlations and low top-20% overlaps show that nodes with the largest GBBRPM increases generally differ from those with the largest positive hydraulic worsening under this protocol. Approximately 71.20% of positive maximum-depth worsening and 40.53% of positive flooding-volume worsening occurred outside the strictly downstream GBBRPM region. This behavior is consistent with backwater, surcharge, ponding, and flow redistribution, which are absent from the downstream-only DAG formulation.

Earlier thesis material preserved historical rank-correlation and top-three values under a different reporting convention. Those values are not used as principal regenerated evidence. The reported top-20% overlap and outside-scope shares are reproduced by the explicit current workflow. Scenario-level and summary outputs are stored in `data/swmm/extracted/swmm_comparison_by_scenario.csv` and `swmm_comparison_summary.csv`.

The mismatch clarifies that GBBRPM v1 is not a surrogate for full hydraulic state prediction. It is better positioned as a lightweight, interpretable downstream comparative-risk propagation and prioritization model.

4.10 Software-Dependency Case

The real Express 4.18.2 dependency topology contains 71 packages and 128 directed dependency edges. Seven package versions were linked to evidence-backed vulnerability advisories and assigned bounded local disturbance levels according to the declared severity mapping.

Table 4.16. Evidence-backed local disturbances in the Express dependency case.

Package	Version	Relation	Severity	B_i	Advisories
body-parser	1.20.1	DIRECT	High	0.75	2
path-to-regexp	0.1.7	DIRECT	High	0.75	3
express	4.18.2	SELF	Moderate	0.50	2
qs	6.11.0	DIRECT	Moderate	0.50	3
cookie	0.5.0	DIRECT	Low	0.25	1
send	0.18.0	DIRECT	Low	0.25	1
serve-static	1.15.0	DIRECT	Low	0.25	1

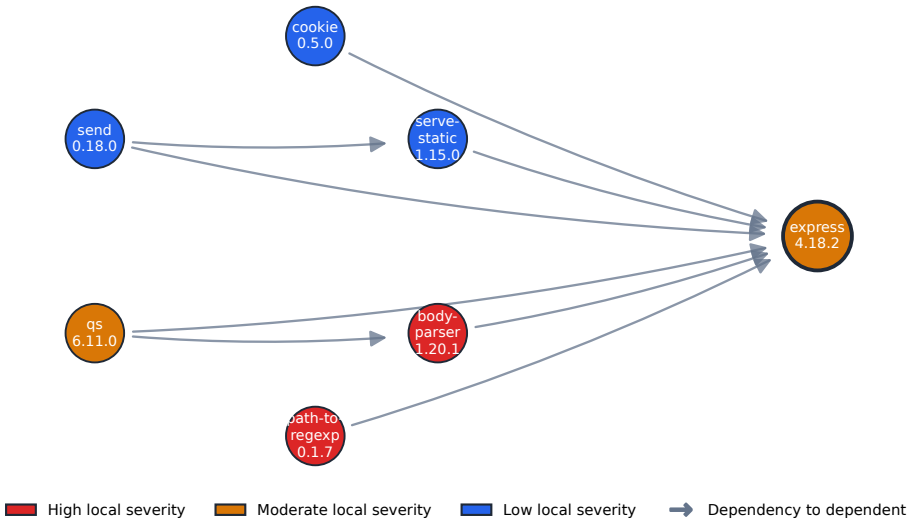


Figure 4.10. Evidence-backed vulnerable packages and their directed dependency relations to the Express root.

The transmission sweep shows that the Express root risk rises from 0.5000 at $\tau = 0$ to 0.9975 at $\tau = 1$. Express moves from rank three to rank one by $\tau = 0.50$, while the complete ordering remains highly stable relative to the $\tau = 1$ reference.

Table 4.17. Software-domain transmission sensitivity and ranking stability.

τ	Express R	Rank	Maximum R	Spearman	Top-20% Jaccard
0.00	0.5000	3	0.7500	0.9993	1.0000
0.25	0.7673	2	0.7812	0.9999	1.0000
0.50	0.9118	1	0.9118	1.0000	1.0000
0.75	0.9766	1	0.9766	1.0000	1.0000
1.00	0.9975	1	0.9975	1.0000	1.0000

Spearman and Jaccard are calculated against the complete ranking at $\tau = 1$.

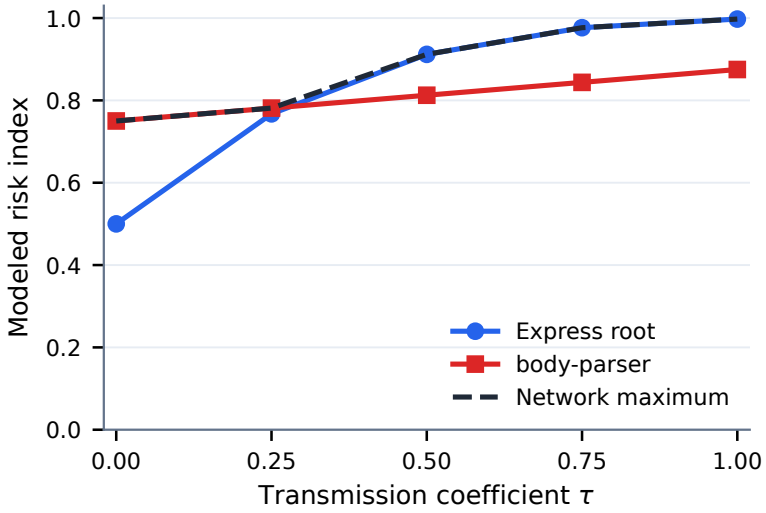


Figure 4.11. Modeled software risk across the transmission-coefficient sweep.

This case demonstrates that the same bounded recursive core can operate on a real directed dependency topology with a domain-specific disturbance mapping. It supports structural applicability and parameter sensitivity, not a claim that the risk index predicts future exploitation or software incidents.

4.11 Electrical-Network Case

4.11.1 Full-Network Import and Baseline Quality

The electrical importer produced 187 bus records and 177 active directed connections distributed across 10 weak components. The baseline archive contained 269 channel files, of which 231 were populated and 38 were empty. Empty members were recorded as data-quality outcomes rather than treated as parser failures.

Table 4.18. Electrical topology and baseline-measurement summary.

Measure	Value
Bus records	187
Active directed connections	177
Weak components (all buses)	10
Documented status transitions	34
Baseline channel files	269
Mapped channel files	222
Populated / empty files	231 / 38
Overall timestamp coverage	98.31%
Usable voltage channels	54
Mean-voltage range	0.9762–1.0431 p.u.

The baseline covers normal operation from November 14–15, 2024 and is not event-response validation.

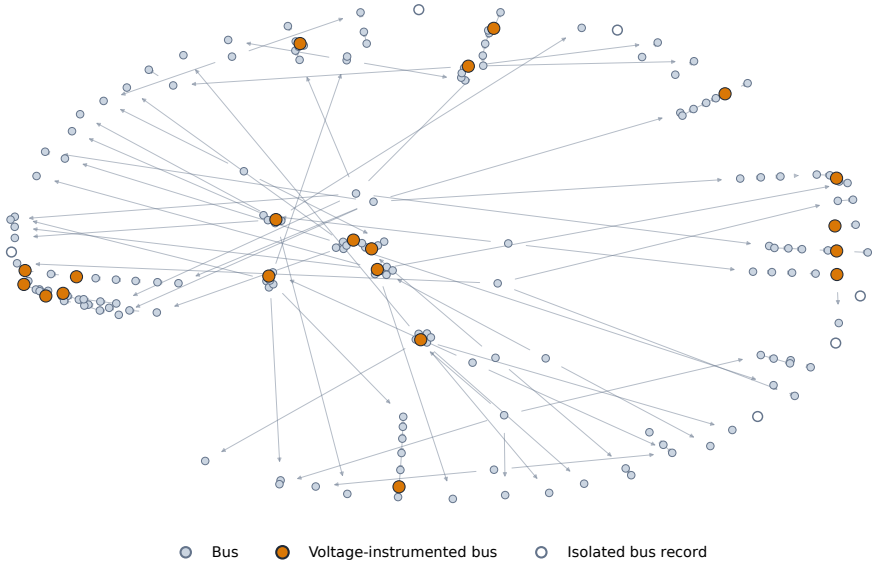


Figure 4.12. Imported electrical topology with voltage-instrumented and isolated bus records identified.

Overall timestamp coverage was 98.31%. The archive contained 54 usable voltage channels with mean values from 0.9762 to 1.0431 p.u. Availability and missing intervals varied across measurement kinds, and 47 channel files remained unmapped under the available meter dictionary.

Table 4.19. Electrical baseline channel availability and timestamp gaps.

Measurement kind	Files	Populated	Empty	Missing intervals
Current Magnitude	119	98	21	20991
Real Power	37	30	7	6997
Unmapped	47	43	4	304520
Voltage Magnitude	66	60	6	5253

Missing intervals are counted at each populated channel's inferred one-second cadence; empty channels are reported separately.

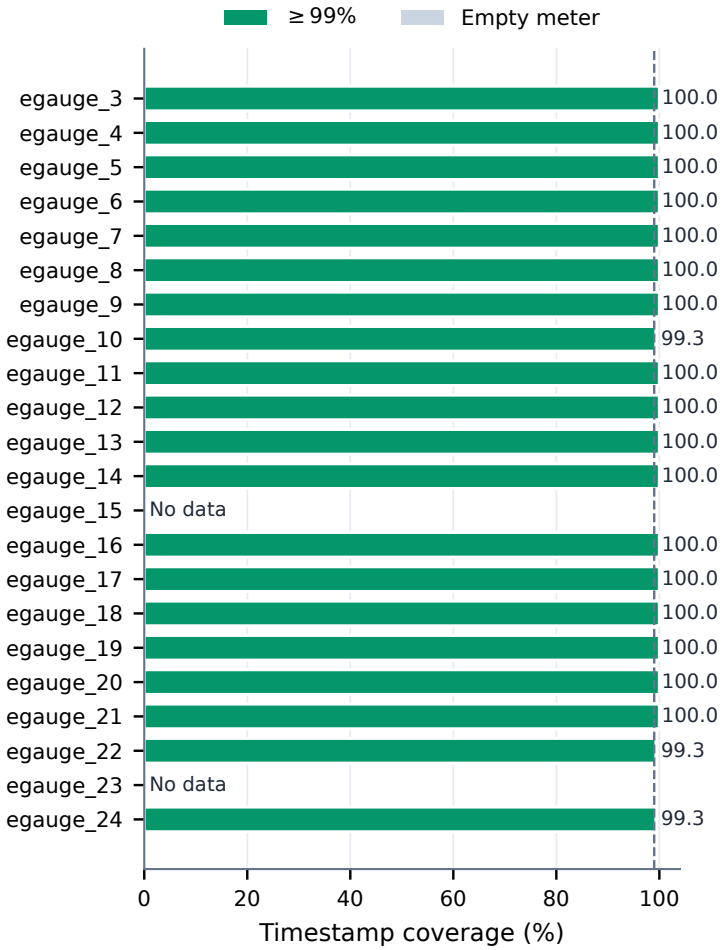


Figure 4.13. Meter-level voltage timestamp coverage during the available normal-operation baseline.

Two meters shown as "No data" in Figure 4.13 lack usable voltage coverage; this does not imply that every mapped current or power channel for those meters is absent. The requested event windows are listed separately because the available baseline covers November 14–15, 2024 rather than the November 13 switching events.

Table 4.20. Requested November 13, 2024 electrical event windows.

Group	Event start	Operations	Kinds	Query start	Query end
E09	17:52:20	4	closing, opening	17:47:20	17:57:28
E10	23:09:00	1	closing	23:04:00	23:14:00
E11	23:20:05	2	closing, opening	23:15:05	23:25:21
E12	23:30:42	1	closing	23:25:42	23:35:42

Event-aligned inputs and independent outcomes were not available in the analyzed extract; these requested windows do not imply completed outcome validation.

This first stage therefore verifies topology construction, channel mapping, chunked baseline preprocessing, empty-file handling, and coverage reporting.

4.11.2 Component 2 Controlled Instantiation

The second stage instantiated GBBRPM on an authorized Component 2 extract from the Caltech SoCal 28-Bus dataset. Table 4.21 summarizes the traceable conversion from topology and phasors to the observable model. The complete 13-node, 12-edge component became a 12-node, 11-edge arborescence after excluding the unobserved `line_407`. The source and five terminal meters had 24 aligned phasor snapshots. Source mean real power was 985.957 kW, compared with 976.828 kW across the five metered terminal loads; the 9.129 kW residual (0.926% of source real power) was retained as an explicit reconciliation residual rather than assigned to an arbitrary node.

Table 4.21. Caltech Component 2 data preparation and model instantiation.

Item	Component 2 value or policy
Complete / observable graph	13 nodes and 12 edges / 12 nodes and 11 edges
Observation exclusion	`line_407`; no supported downstream meter path
Measurement window	24 synchronized phasor snapshots from November 14, 2024 07:00
Source / terminal meters	`egauge_21` / `egauge_3, 7, 9, 11, 13`
Mean real power	985.957 kW source; 976.828 kW metered terminal loads
Unallocated residual	9.129 kW and 135.398 kvar, reported separately
Susceptibility	$S_{ij} = \min(1, L_{ij}/C_{ij})$ using mean downstream apparent power
Capacity	Declared line current and voltage; minimum transformer nameplate kVA
Reference state	$B_i = 0$ and $\tau_{ij} = 1$ unless varied by the scenario

Phasor-derived power is the model input. The raw `Mains_Power` register is retained only as a scale cross-check because its declared unit is inconsistent with its numerical values.

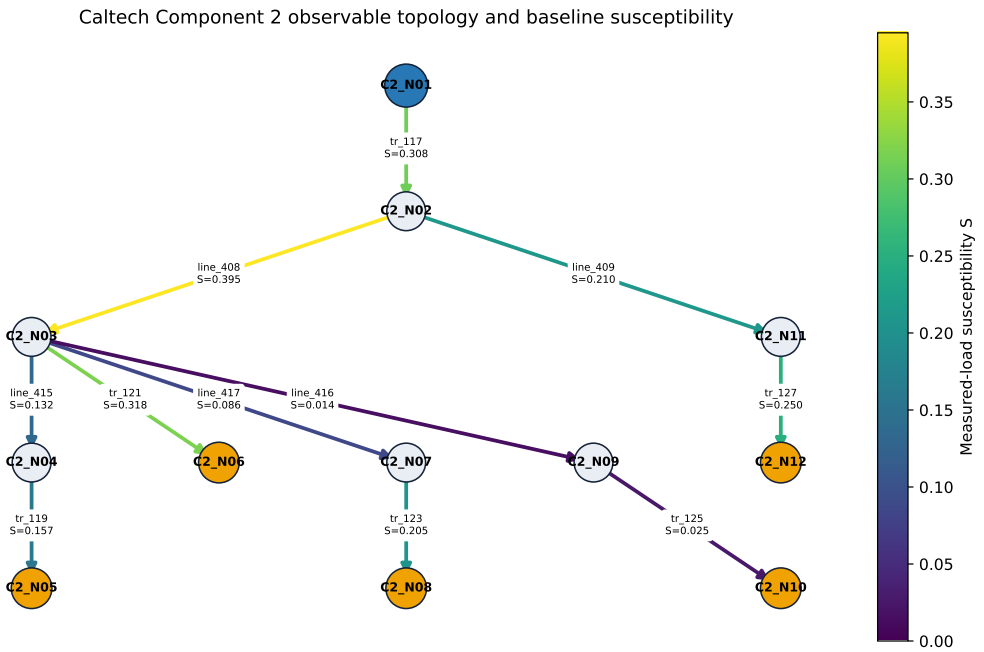


Figure 4.14. Observable Component 2 topology and baseline edge susceptibility derived from measured downstream apparent power and declared capacity.

Figure 4.14 shows that measured loading produces heterogeneous electrical susceptibility even though the graph is small. The largest baseline susceptibility is 0.395 on `line_408`, followed by 0.318 on transformer `tr_121` and 0.308 on source transformer `tr_117`; the lightly loaded `line_416` branch has susceptibility 0.014. These values are descriptive loading-to-capacity ratios, not empirical failure probabilities.

All 43 controlled runs preserved node risks and edge contributions in $[0, 1]$. The zero-disturbance reference produced zero risk throughout. With a root disturbance of 0.75, the risk sum was 1.186, three nodes exceeded 0.05, and the maximum terminal-node risk was 0.029. Raising the root disturbance from 0.25 to 1.00 increased risk sum monotonically from 0.395 to 1.581 and increased the number of affected nodes from two to four. Table 4.22 reports selected endpoints from all major scenario families.

Table 4.22. Selected Component 2 controlled-scenario results.

Scenario	Setting	Risk sum	Max. risk	Affected	Highest-risk node
Reference	$B = 0$	0.000	0.000	0	None
Root severity	$B_{N01} = 0.25$	0.395	0.250	2	C2_N01
Root severity	$B_{N01} = 0.75$	1.186	0.750	3	C2_N01
Root severity	$B_{N01} = 1.00$	1.581	1.000	4	C2_N01
Load scale	$0.50\times$	0.908	0.750	2	C2_N01
Load scale	$1.50\times$	1.638	0.750	5	C2_N01
Transmission	$\tau = 0$	0.750	0.750	1	C2_N01
Transmission	$\tau = 1$	1.186	0.750	3	C2_N01
Location	$B_{N02} = 0.75$	1.417	0.750	4	C2_N02
Location	$B_{N03} = 0.75$	1.191	0.750	4	C2_N03
All internal	six nodes at $B = 0.50$	3.754	0.599	10	C2_N03

"Affected" denotes observable nodes with risk greater than 0.05. Values are deterministic comparative indices, not probabilities or observed event outcomes.

Uniformly scaling measured edge loads while holding the root disturbance at 0.75 produced a monotone increase in propagated risk. Risk sum rose from 0.908 at $0.50\times$ measured loading to 1.638 at $1.50\times$, while the affected-node count rose from two to five. The maximum risk remained 0.75 because the controlled source disturbance was fixed; the change occurred in downstream propagation rather than in the injected source value.

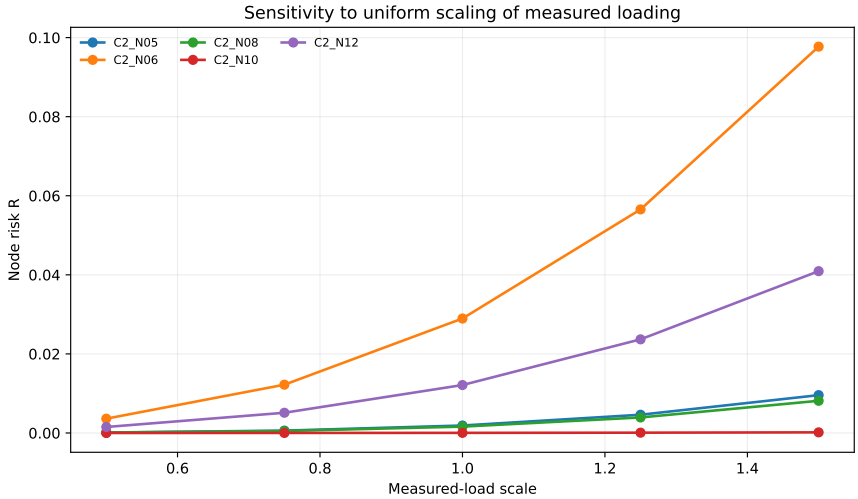


Figure 4.15. Component 2 response to uniform counterfactual scaling of measured edge loading with root disturbance fixed at 0.75.

Disturbance placement also changed total propagated risk. At equal local severity 0.75, C2_N02 produced the largest risk sum (1.417), followed by C2_N03 (1.191) and the root C2_N01 (1.186). C2_N02 avoids the initial attenuation through `tr_117` while retaining access to nearly all downstream branches. Terminal-node disturbances produced no downstream propagation and therefore retained a risk sum of 0.750. This result shows why equal local disturbances need not imply equal network-level priorities.

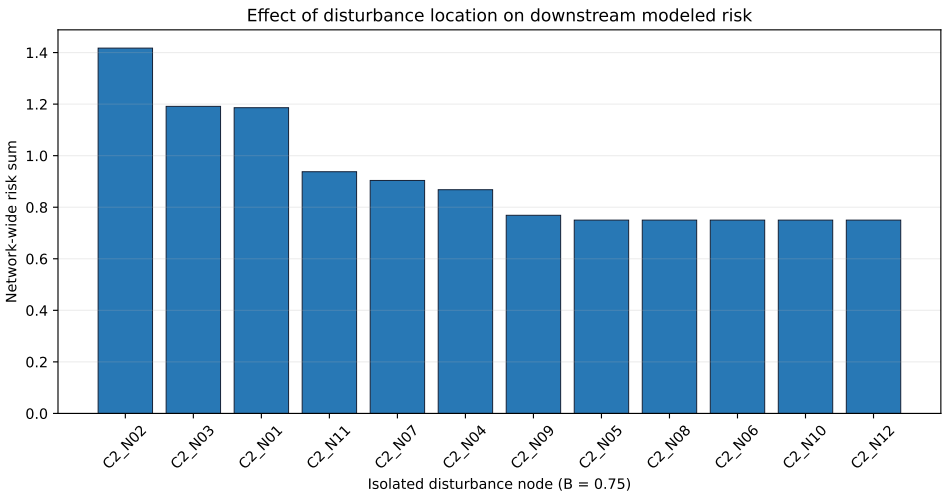


Figure 4.16. Risk sum under an isolated disturbance of 0.75 at each observable Component 2 node.

The uniform transmission sweep provided a second controlled attenuation test. With $\tau = 0$, only the root retained its local risk and the risk sum was 0.750. Increasing τ to 1 raised risk sum monotonically to 1.186 and maximum terminal risk to 0.029. In the multi-source family, the nested C2_N02–C2_N03 pair produced a maximum node risk of 0.599 and risk sum of 1.582, demonstrating bounded combination of incoming and local disturbance. Activating all six internal non-root sources at 0.50 raised risk sum to 3.754 and affected 10 of 12 nodes. Parallel-source cases represent concurrent branch coverage; because Component 2 is an arborescence, these runs are not evidence about reconvergent-path dependence.

The Component 2 results extend the electrical case beyond importer and baseline verification to controlled mechanism and sensitivity evidence on real topology and measured normal-operation loading. They still do not validate GBBRPM against observed electrical-event outcomes. Such a claim requires temporally aligned disturbance labels and independent outcome measurements for the requested switching windows.

4.12 Integrated Discussion

Taken together, the experiments support five principal conclusions. First, GBBRPM satisfies its declared bounded and monotone risk-index semantics under the tested conditions. Second, graph structure and heterogeneous susceptibility materially influence downstream ranking beyond local-only scoring. Third, prioritization is generally stable under small perturbations but exact top-ranked membership becomes less stable under larger uncertainty. Fourth, the SWMM comparison demonstrates that the model's value lies in lightweight interpretable propagation and prioritization rather than high-fidelity hydraulic reproduction. Fifth, the software and electrical cases show that the frozen architecture and data pipeline can be mapped to distinct real directed-network structures, although their evidence does not establish equal predictive validity. The Component 2 suite specifically shows that measured electrical loading can parameterize het-

erogeneous susceptibility and produce expected bounded, monotone scenario responses without converting those responses into event predictions.

The results support GBBRPM as an integrative graph-risk architecture rather than a claim that any individual mechanism is novel. They also identify extension targets including calibrated transmission, dependence-aware aggregation, cyclic or bidirectional propagation, and application-specific parameter estimation.

4.13 Chapter Summary

The restored pipeline provides a coherent computational assessment of GBBRPM v1 across five controlled synthetic DAGs, 136 controlled scenario runs, diagnostic comparators, 600-trial robustness experiments, randomized property tests, reconvergence diagnostics, scalability tests, and a separate SWMM external-reference workflow. Cross-domain evidence adds a 71-package software dependency graph, a 187-bus electrical baseline with 269 channel files, and a 12-node observable Component 2 model evaluated in 43 controlled runs.

The evidence supports bounded recursive propagation, sensitivity to network context, robustness under modest uncertainty, and computational tractability while making the limitations of shared-path dependence and downstream-only propagation explicit. They also support cross-domain instantiability within the tested evidence boundaries while keeping electrical event validation pending. These findings provide the basis for the conclusions and recommendations in the succeeding chapter.

CHAPTER V
SUMMARY AND CONCLUSION

5.1 Overview

This study designed and evaluated GBBRPM v1 as a bounded recursive model for comparative node-risk prioritization in directed acyclic networks. The evidence combines controlled architecture tests with three domain-specific cases: drainage, software dependency, and electrical networks. The cases intentionally support different claims and are not presented as equivalent forms of outcome validation.

5.2 Summary of Findings

5.2.1 Architecture Behavior

Across the controlled N1–N5 experiments, GBBRPM preserved the declared $[0, 1]$ bounds and monotone response to increases in local disturbance and susceptibility. Recursive current-state propagation, heterogeneous edge conditions, disturbance location, and multi-source configuration materially affected both risk magnitude and ranking. Randomized property tests passed all 600 trials for boundedness, monotonicity, full local disturbance, and zero-susceptibility gating.

Ranking robustness remained high under modest perturbations, although agreement among the highest-priority nodes declined faster than the overall Spearman ordering. The reconvergence diagnostic also showed that shared ancestry can be counted through multiple paths, especially in N5. This is a known boundary of the local multiplicative-complement aggregator rather than evidence of probabilistic independence.

The sparse-DAG timing experiment was consistent with the expected one-pass $O(|V| + |E|)$ implementation. Median execution time increased from approximately 1.63 ms at 20 nodes to 16.65 ms at 1000 nodes under the reported environment.

5.2.2 Drainage Evidence and Hydraulic Boundary

The drainage case supplied the most complete behavioral evaluation. The controlled experiments demonstrated the effects of attenuation, branching, convergence, reconvergence, local disturbance, susceptibility, and multiple active sources. However, the 12-scenario SWMM comparison produced negative mean rank association and low top-20% overlap for both maximum-depth and flooding outcomes. Hydraulic worsening also occurred outside the model's strictly downstream region.

These findings do not support interpreting the GBBRPM index as a hydraulic predictor. Instead, they identify a clear scope boundary: GBBRPM

represents lightweight downstream comparative propagation, whereas SWMM represents hydraulic mechanisms such as backwater, surcharge, ponding, and flow redistribution.

5.2.3 Software-Dependency Evidence

The Express 4.18.2 case instantiated the frozen core on a real topology of 71 packages and 128 dependency edges, with seven evidence-backed affected versions. Express root risk increased from 0.5000 to 0.9975 across the transmission sweep and reached rank one by $\tau = 0.50$. Ranking agreement with the $\tau = 1$ reference remained high.

This result supports real-topology structural applicability and expected sensitivity to transmission. It does not establish that the modeled index is an exploit probability or a predictor of future incidents.

5.2.4 Electrical-Network Evidence

The full-network electrical stage verified import and baseline preprocessing for 187 bus records, 177 active directed connections, and 269 channel files. The workflow handled 38 empty archive members without terminating, reported mapping and missingness explicitly, identified 54 usable voltage channels, and measured 98.31% overall timestamp coverage.

The Component 2 stage extended this evidence through a controlled real-topology, measured-loading instantiation. Its observable model contained 12 nodes and 11 edges parameterized from 24 aligned phasor snapshots. Across 43 deterministic scenarios, risks and edge contributions remained bounded. Root severity, measured-load scaling, and uniform transmission produced monotone increases in downstream risk. Equal disturbances at different nodes produced different risk sums, with C2_N02 highest at 1.417 because it bypassed the initial transformer attenuation while retaining broad downstream reach. The all-internal-source case affected 10 of 12 nodes while remaining bounded at a maximum node risk of 0.599.

These findings support controlled electrical mechanism and sensitivity claims, not event prediction. The measurements represent normal operation and are not aligned with the requested switching-event outcomes; electrical event-response validation therefore remains pending.

5.3 Answers to the Research Questions

The combined evidence answers the five research questions within the declared non-probabilistic and non-predictive scope:

RQ1. Local disturbance, current upstream risk, relation-specific susceptibility, optional transmission, directed reachability, and multi-source input are integrated through a topologically ordered evaluator and the multiplicative-complement aggregation rule. The resulting state remains traceable to its local and incoming terms.

RQ2. The formulation preserved the declared $[0, 1]$ bounds and monotone re-

sponse in the controlled scenarios and all 600 randomized property trials. Comparative rankings also changed in the expected direction when source severity, edge susceptibility, and disturbance placement changed.

RQ3. The model responded materially to severity, susceptibility, topology, location, and multi-source configuration; retained high overall ranking agreement under modest perturbation; exposed reduced top-set stability at larger perturbations; and exhibited runtime growth consistent with one-pass $O(|V| + |E|)$ evaluation on the tested sparse DAGs.

RQ4. The frozen update rule was instantiated without redefinition in drainage, software-dependency, and electrical-network cases. The domain meanings and evidence sources changed, but the roles of B, S, τ, Q, and R remained fixed.

RQ5. The domains provide unequal but complementary evidence: drainage supplies controlled behavior and a hydraulic-reference boundary; software supplies real-topology structural and transmission-sensitivity evidence; and electrical data supply importer verification plus controlled real-topology, measured-loading sensitivity evidence. None of these results establishes universal predictive validity, and the electrical case still lacks event-aligned independent outcomes.

5.4 Conclusions

Within the tested scope, the study concludes that:

1. GBBRPM v1 provides a deterministic, bounded, and interpretable recursive architecture for comparative risk prioritization on DAGs.

2. Network structure and heterogeneous relation conditions can change prioritization beyond what local disturbance alone indicates.

3. The model is broadly robust to modest input perturbations, but exact membership of the highest-priority set becomes less stable as uncertainty increases.

4. The same frozen propagation rule can be instantiated coherently in the tested drainage, software, and electrical network structures. This supports cross-domain instantiability, not universal validity.

5. The SWMM disagreement establishes that GBBRPM should not be used as a substitute for hydraulic simulation.

6. The electrical evidence verifies topology and baseline preprocessing and demonstrates bounded, monotone controlled responses on Component 2 under measured loading. Event-centered validation cannot be claimed until temporally aligned inputs and independent outcomes are obtained.

5.5 Limitations

GBBRPM v1 is limited to directed acyclic graphs and one-pass propagation. It does not model cycles, feedback, bidirectional influence, explicit time evolution, or common-cause dependence correction. The bounded risk state is a comparative index rather than a calibrated probability. Domain-specific disturbance, susceptibility, and transmission mappings remain dependent on the quality and defensibility of their input evidence.

The drainage case uses controlled networks and simulated hydraulic reference outcomes rather than a calibrated operational field study. The software case demonstrates structural applicability and sensitivity rather than observed incident prediction. The Component 2 electrical suite injects controlled disturbances into a measured normal-operation baseline; it does not contain observed event labels or outcomes aligned with the requested windows. Component 2 was selected purposively using observability, traceability, branching structure, and DAG compatibility, not through random sampling; the 24-snapshot window supports descriptive parameter instantiation rather than temporal or population-level generalization. Its residual power is reported but not spatially allocated, and one unobserved component edge is excluded. The three cases therefore cannot be pooled as if they offered identical empirical strength.

CHAPTER VI
RECOMMENDATIONS

The following recommendations follow directly from the evidence limits identified in the drainage, software-dependency, and electrical-network cases:

1. Obtain electrical measurements and independent event outcomes aligned with the four requested November 13 event groups, then execute the preregistered event-window analysis against the Component 2 controlled baseline.
2. Evaluate domain-specific rankings against independent outcomes that were not used to construct the model inputs.
3. Obtain provenance-complete operational drainage observations so that the controlled and SWMM-based findings can be compared with field conditions rather than being interpreted as field validation.
4. Investigate ancestry-aware or dependence-aware aggregation for reconvergent paths while retaining the current v1 formulation as the comparison baseline.
5. Calibrate or justify heterogeneous transmission and susceptibility functions separately within each domain.
6. Develop explicit extensions for cyclic, bidirectional, and temporal networks rather than silently applying the DAG evaluator outside its assumptions.
7. Repeat the cross-domain protocol on additional datasets to test out-of-sample applicability without overstating generalizability.

These recommendations preserve the distinction between the model's validated computational behavior and the empirical claims that still require domain-specific evidence. GBBRPM v1 is not a universal predictor, a hydraulic simulator, or a completed electrical event model; subsequent work should retain those interpretation boundaries while strengthening field evidence and dependence handling.

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APPENDIX A

Worked Example of the Generic GBBRPM Core Architecture

A.1 Purpose and Scope of the Example

This appendix provides a transparent, step-by-step calculation of the generic Graph-Based Bounded Risk Propagation Model (GBBRPM) core architecture. The example demonstrates source-node initialization, relation-level contribution, bounded convergence, local disturbance at a receiving node, current-state recursion, and final comparative ranking. All values are controlled, dimensionless illustrative inputs. They are not drainage measurements, software observations, electrical measurements, calibrated probabilities, or evidence of empirical predictive validity.

The calculation uses the controlled directed acyclic graph shown in Figure A.1. Its directed relations are $N_1 \to N_2$, $N_4 \to N_2$, and $N_2 \to N_3$. Nodes N_1 and N_4 are sources, the two incoming paths converge at N_2, and the accumulated state at N_2 is then propagated to N_3. A valid topological evaluation order is N_1, N_4, N_2, N_3.

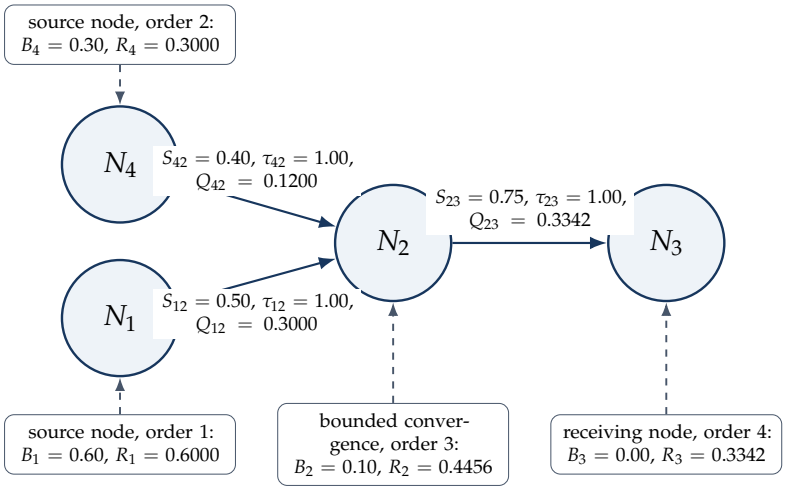


Figure A.1. Controlled four-node DAG used in the Appendix A calculation. Node callouts show evaluation order, local disturbance, and computed state; relation labels show susceptibility, neutral transmission, and incoming contribution.

A.2 Controlled Inputs

Table A.1 lists the local disturbance assigned to each node. The nonzero B_2 value is included deliberately to show that a receiving node can contain both locally

introduced and inherited risk.

Table A.1. Controlled Node Inputs for the Worked Example

Node	Predecessor set $P(j)$	Local disturbance B_j
N_1	$\emptyset$	0.60
N_4	$\emptyset$	0.30
N_2	$\{N_1, N_4\}$	0.10
N_3	$\{N_2\}$	0.00

Table A.2 lists the relation-specific susceptibilities and transmission factors. Neutral transmission $\tau_{ij} = 1$ is used consistently with the frozen v1 baseline. Therefore, the worked example does not claim to evaluate heterogeneous transmission.

Table A.2. Controlled Relation Inputs for the Worked Example

Relation	Susceptibility S_{ij}	Transmission τ_{ij}	Role
$N_1 \rightarrow N_2$	0.50	1.00	First input to convergence
$N_4 \rightarrow N_2$	0.40	1.00	Second input to convergence
$N_2 \rightarrow N_3$	0.75	1.00	Recursive downstream transfer

All supplied values lie in $[0, 1]$, every relation has a valid endpoint, and the graph contains no directed cycle. The inputs therefore satisfy the declared bounded-input and DAG requirements before propagation begins.

A.3 Core Update Rules

For a directed relation $i \rightarrow j$, the incoming contribution is calculated as

$$Q_{ij} = S_{ij}\tau_{ij}R_i. \qquad \text{(Equation A-1)}$$

where:
Q_{ij} = contribution entering node j from predecessor i;
S_{ij} = bounded relation-specific susceptibility;
τ_{ij} = bounded transmission or influence factor; and
R_i = current accumulated state of upstream node i.

The receiving-node state is calculated as

$$R_j = 1 - (1 - B_j) \prod_{i \in P(j)} (1 - Q_{ij}). \qquad \text{(Equation A-2)}$$

where:
R_j = final bounded state of receiving node j;
B_j = local disturbance introduced directly at node j;

$P(j)$ = set of immediate predecessors of node j; and
Q_{ij} = incoming contribution from predecessor i.

The multiplicative-complement expression is a bounded saturating aggregator. It is used algebraically and is not interpreted as a probabilistic union.

A.4 Step 1: Initialize the Source Nodes

Nodes N_1 and N_4 have no predecessors. For either source node, the product over the empty predecessor set equals one. Equation A-2 therefore reduces to $R_j = B_j$:

$$
\begin{aligned}
R_1 &= 1 - (1 - B_1) \prod_{i \in \emptyset} (1 - Q_{i1}) \\
&= 1 - (1 - 0.60)(1) \\
&= 0.60, \\
R_4 &= 1 - (1 - B_4) \prod_{i \in \emptyset} (1 - Q_{i4}) \\
&= 1 - (1 - 0.30)(1) \\
&= 0.30.
\end{aligned}
\qquad \text{(Equation A-3)}
$$

Thus, a source node can contain a local disturbance. It does not require a special propagation rule: the ordinary aggregation rule produces its local state automatically through the empty-product convention.

A.5 Step 2: Calculate Contributions Entering N_2

The current source states are now available for the two relations entering N_2. Applying Equation A-1 gives

$$
\begin{aligned}
Q_{12} &= S_{12} \tau_{12} R_1 \\
&= (0.50)(1.00)(0.60) \\
&= 0.30, \\
Q_{42} &= S_{42} \tau_{42} R_4 \\
&= (0.40)(1.00)(0.30) \\
&= 0.12.
\end{aligned}
\qquad \text{(Equation A-4)}
$$

The two contributions are calculated separately because they arrive through different directed relations with different susceptibilities. Their raw sum is not used as the receiving-node state; they are combined through the bounded aggregation rule in the next step.

A.6 Step 3: Aggregate Local and Inherited Risk at N_2

Node N_2 has local disturbance $B_2 = 0.10$ and two incoming contributions. Substitution into Equation A-2 gives

$$\begin{aligned} R_2 &= 1 - (1 - B_2)(1 - Q_{12})(1 - Q_{42}) \\ &= 1 - (1 - 0.10)(1 - 0.30)(1 - 0.12) \\ &= 1 - (0.90)(0.70)(0.88) \\ &= 1 - 0.5544 \\ &= 0.4456. \end{aligned} \qquad \text{(Equation A-5)}$$

The value $R_2 = 0.4456$ represents the combined effect of the local disturbance at N_2 and the two inherited contributions. Although all three components contribute to the result, the multiplicative-complement form keeps the computed state within $[0, 1]$.

For comparison, removing the local disturbance while retaining both inherited contributions would give $1 - (1 - 0.30)(1 - 0.12) = 0.3840$. The increase from 0.3840 to 0.4456 is therefore caused by including the receiving node's own local disturbance through the same bounded rule rather than adding it after propagation.

A.7 Step 4: Propagate the Current State to N_3

The contribution from N_2 to N_3 uses the newly accumulated state $R_2 = 0.4456$, not only the original local disturbance $B_2 = 0.10$:

$$\begin{aligned} Q_{23} &= S_{23}\tau_{23}R_2 \\ &= (0.75)(1.00)(0.4456) \\ &= 0.3342. \end{aligned} \qquad \text{(Equation A-6)}$$

Because N_3 has no local disturbance and only N_2 as its predecessor, its state is

$$\begin{aligned} R_3 &= 1 - (1 - B_3)(1 - Q_{23}) \\ &= 1 - (1 - 0.00)(1 - 0.3342) \\ &= 1 - (1.00)(0.6658) \\ &= 0.3342. \end{aligned} \qquad \text{(Equation A-7)}$$

This step demonstrates current-state recursion. If the calculation incorrectly used only B_2, the relation contribution would be $(0.75)(1.00)(0.10) = 0.0750$, which would discard the risk inherited by N_2. The implemented architecture instead propagates R_2, preserving the accumulated upstream effect in the downstream evaluation.

A.8 Step 5: Produce the Comparative Ranking

After all nodes have been evaluated once in topological order, their final states are sorted in descending order. Table A.3 reports the resulting prioritization.

Table A.3. Final Node States and Comparative Ranking

Rank	Node	Final state R_i
1	N_1	0.6000
2	N_2	0.4456
3	N_3	0.3342
4	N_4	0.3000

The ranking is a comparison among modeled node states under this controlled configuration. It does not mean that N_1 has a 60% probability of failure or that the numerical spacing between ranks has a domain-independent physical meaning. If two nodes obtain exactly equal final states, the implementation retains the tie and assigns them the same risk standing rather than applying an unsupported secondary tie-breaker.

A.9 Calculation Trace and Interpretation

Table A.4 consolidates the complete evaluation trace. It also shows that every downstream calculation uses only predecessor states that have already been computed in the declared topological order.

Table A.4. Complete Topological Evaluation Trace

Order	Node	Available predecessor states	Incoming contributions	Result
1	N_1	None	Empty product	$R_1 = 0.6000$
2	N_4	None	Empty product	$R_4 = 0.3000$
3	N_2	$R_1 = 0.6000$, $R_4 = 0.3000$	$Q_{12} = 0.3000$, $Q_{42} = 0.1200$	$R_2 = 0.4456$
4	N_3	$R_2 = 0.4456$	$Q_{23} = 0.3342$	$R_3 = 0.3342$

The worked example establishes seven features of the generic architecture. First, the empty-product convention gives source nodes their local states without a separate initialization equation. Second, relation susceptibility modifies each upstream state before it enters the receiving node. Third, neutral transmission preserves the transmission interface without introducing unsupported relation differences. Fourth, multiple incoming contributions and a receiving-node disturbance are combined through one bounded rule. Fifth, the evaluator propagates the current accumulated state rather than only the original local input. Sixth, one topological pass is sufficient because the evaluated graph is acyclic. Seventh, prioritization is derived only after all node states have been calculated.

These calculations explain the mechanics of the frozen core architecture. They do not establish that the illustrative parameter values are appropriate for a particular domain. Drainage, software-dependency, and electrical-network instantiations require separate evidence for the meanings and construction of their local disturbances, susceptibilities, transmission factors, and outcome comparators.